Energy Nexus 17 (2025) 100373

Contents lists available at ScienceDirect

Energy Nexus

journal homepage: www.elsevier.com/locate/nexus

Emerging technologies in water desalination: A review and future outlook

Anwur Alenezi [a,*], Yousef Alabaiadly [b]

[a] Water Research Centre, Kuwait Institute for Scientific Research, PO Box 24885, Safat 13109, Kuwait
[b] Studies and Research Department, Ministry of Electricity and Water, Ministries Zone 12010, Kuwait

ARTICLE INFO

Keywords:
Desalination
Solar electrochemical distillation
Capacitive deionisation
Membrane distillation
Forward osmosis
Water scarcity

ABSTRACT

This review analyses emerging desalination technologies that offer sustainable solutions to global water scarcity and address unresolved issues. This study examines solar electrochemical distillation (SED), integrated solar capacitive deionisation (SCDI) and capacitive deionisation hybrid systems (CDI-HS), solar-powered passive desalination (SPPD), membrane distillation (MD), and forward osmosis (FO). A comprehensive literature review evaluates recent research and advancements in each technology. From each study, the authors extracted key characteristics, including the year of publication, research methods, data collection techniques, and the direction or strength of the research outcomes. Each study in this review serves as a unit of analysis, with the literature forming a database to interpret trends and draw conclusions about emerging desalination technologies. Key challenges were identified, and recommendations for future studies proposed, based on existing data and experimental findings. The review's findings underscore the need to address unresolved issues in desalination technologies to enhance their efficiency, scalability, and sustainability. Implementing the recommended research strategies could optimise these technologies, ensuring a reliable and sustainable supply of fresh water. Continued innovation, supported by targeted research and robust policy frameworks, is essential to mitigate global water scarcity and ensure water security for future generations.

1. Introduction

Water is a vital resource for life and plays a significant role in global social, economic, and environmental development. Currently, water is abundant yet unevenly distributed [1]. Water covers 70 % of our planet's surface, leading to the misconception that it is plentiful. However, access to fresh water is limited [2]. Water scarcity is a global issue that affects even countries with sufficient water resources. Globally, approximately 1.1 billion people lack access to clean water, while a further 2.7 billion face water scarcity for at least one month each year. Considering the current rate of water consumption, this situation is likely to worsen in the coming years. After 2030, nearly two-thirds of the global population could face water shortages, resulting in further suffering for ecosystems worldwide. Further predictions based on World Bank probable population data and the FAO AQUASTAT database indicate that by 2050, two billion people in 44 countries could experience water scarcity. Water scarcity threatens health and livelihoods and poses significant challenges to economic development and social stability, especially in vulnerable regions.

Desalination presents a viable solution to freshwater scarcity, particularly in developing nations. Desalination is often described as a drought-proof water source, as it does not depend on river flow, reservoir levels, or climate change [3]. This method is particularly useful in addressing scarcity in industrial and coastal cities. Therefore, desalination could be widely adopted to address water scarcity. Although desalination is primarily used to produce fresh water from seawater, it can also treat saline or brackish water, low-grade surface water, groundwater and treated effluents [3]. Recent advancements in desalination technologies have led to improved efficiency and reduced costs, making these methods suitable alternatives for traditional fresh water sources.

Various traditional desalination methods exist categorised into two,

Abbreviations: AGMD, Air-gap membrane distillation; AI, Artificial intelligence; AC, Activated carbon; ANFIS, Adaptive neuro-fuzzy inference system; CDI, Capacitive deionisation; CDI-HS, Capacitive deionisation hybrid systems; DCMD, Direct contact membrane distillation; EDL, Electrical double layer; FFNN, Feed-forward neural network; FO, Forward osmosis; MCDI, Membrane capacitive deionisation; MD, Membrane distillation; PCMs, Phase change materials; SCDI, Integrated solar capacitive deionisation; SED, Solar electrochemical distillation; SGMD, Sweeping gas membrane distillation; SPPD, Solar-powered passive desalination; STEP, Solar thermal electrochemical photo; SVM, Support vector machine; UDDM, Uniform density desalination membranes; VMD, vacuum membrane distillation.

* Corresponding author.
E-mail addresses: anwenezi@kisr.edu.kw (A. Alenezi), yaalobidly@mew.gov.kw (Y. Alabaiadly).

https://doi.org/10.1016/j.nexus.2025.100373
Received 30 August 2024; Received in revised form 13 December 2024; Accepted 7 February 2025
Available online 8 February 2025
2772-4271/© 2025 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

as either membrane or thermal-based desalination processes [4,5]. The key processes in thermal distillation processes include heating, evaporation, and condensation of salt water, and produce high salinity brine and distilled freshwater [4,6,7]. Among thermal distillation methods, Multi-Stage Flash (MSF), Multi-Effect Distillation (MED) and vapour-compression evaporation are the most common [5]. In MSF distillation, saline water is conveyed through various chambers where it is heated and compressed at high temperatures and pressures. The pressure decreases as the water moves through the chambers, resulting in the boiling of saline water and the formation of fresh water vapour, which is condensed and collected [5]. MSF has a recovery rate of 25–30 % [8]. The approach consumes high quantities of energy due to heating requirements, most of which is thermal energy from power plants. Hence, while MSF is a reliable approach, it has high operational costs related to energy use. MED works by using various stages where seawater is heated and evaporated in successive vessels. The vapour arising from one effect is used to heat the next, improving the energy efficiency of the system. The recovery rate of MED ranges between 10 % and 20 %, depending on the system's design and operating conditions [9]. The energy consumption of MED systems is much lower than MSF because of heat recovery and depends on the number of effects used in the system. MED is popular for its high efficiency in large-scale applications; however, it needs major capital investments. Thermal-based desalination plants integrated with renewable sources of energy or those using waste heat as energy emit less greenhouse gas [10]. Renewable technologies suited to thermal-based desalination plants include solar thermal, solar photovoltaics (PV), wind, and geothermal energy. Solar technologies reliant on solar heat concentration, particularly concentrating solar power (CSP), produce large quantities of heat suited to thermal desalination [11]. Overall, thermal-based desalination processes provide a reliable source of fresh water but are energy intensive. The successful implementation of desalination technologies will require supportive policies and infrastructure investment to overcome economic barriers. However, it is important to consider the sustainability of these technologies, especially regarding brine disposal and energy consumption.

In membrane-based desalination processes, high-pressure saline water flows through a semi-cellular membrane to produce freshwater. Compared to thermal distillation systems, membrane processes consume less energy, are more compact, and result in high modularity [4]. The most common membrane processes are forward osmosis (FO) and reverse osmosis (RO) [4,7,12]. RO predominates the global distillation market and uses a semi-permeable membrane to separate salts from water by applying pressure to the saline feed water. This process allows fresh water to pass through while retaining the salts. The recovery rate for RO ranges between 40 and 60 % for seawater [13]. The energy consumption is much lower compared to the thermal methods. Modern RO systems have energy recovery devices, which are designed for energy efficiency. RO has been adopted widely because of its lower energy requirements and simple operations. However, it requires pre-treatment of feedwater to minimise membrane fouling [4,14]. Membrane-based desalination processes also play a key role in reducing CO_2 emissions. Photovoltaic and wind electricity are commonly combined with membrane desalination units such as reverse osmosis to reduce emissions. While membrane-based desalination processes have a high recovery ratio, they consume a lot of energy in heating and cooling and report a lot of energy losses through the membrane [15].

In desalination systems, numerous sorts of solar stills exist. These devices are cheap, easy to construct, and manageable tools for supplying potable water to rural communities. They use a cheap source of energy – the sun – to offer a reliable solution for water desalination [16,17]. The most common device is the conventional type of solar still, often referred to as the basin solar still [17]. However, the main concern with this device is its low productivity [17]. Researchers have tried to improve the productivity of the conventional solar still through different designs and operating parameters. Modifications have led to advanced designs such as pyramid triangular solar stills, double basins, tubular, hemispherical, and spherical devices [17]. Some researchers also recommend integrated systems, such as connected solar stills with photo voltaic cells (PVC) or solar collectors [17,18]. Fig. 1 is a general schematic representation of solar stills and their basic operation.

This review provides a detailed overview of emerging desalination technologies that are key in addressing the escalating global water scarcity crisis. The demand for clean water is particularly acute in arid and semi-arid regions, where traditional water sources are proving increasingly insufficient. Innovative desalination methods are important to meet the current demands and anticipate future challenges. The review highlights the potential and limitations of these emerging technologies, providing important insights into their drawbacks, such as membrane fouling in RO systems and high energy demands in thermal processes, such as MSF distillation. By identifying these unresolved issues and providing targeted recommendations for future research, this work encourages innovations that enhance the reliability and sustainability of clean water supply systems. Addressing these challenges through progressive innovation and supportive policies is key for ensuring long-term water security in the face of growing global populations and climate change.

2. Newly discovered technologies

2.1. Solar electrochemical distillation

A recently discovered technology with significant potential for water desalination is solar electrochemical distillation (SED). This technology is a promising, energy-efficient and sustainable alternative for water desalination, utilising solar energy to separate fresh water from saline water and other contaminants [19,20]. Unlike traditional solar distillation processes that rely solely on thermal energy, this technique combines solar energy with electrochemical processes for water desalination.

2.1.1. Mechanism of operation

SED integrates solar energy with electrochemical processes to enable desalination (Fig. 2). The system comprises three main components: a solar absorber, an electrochemical cell and a distillation unit [19,20]. The solar absorber collects solar energy to generate heat, which is used

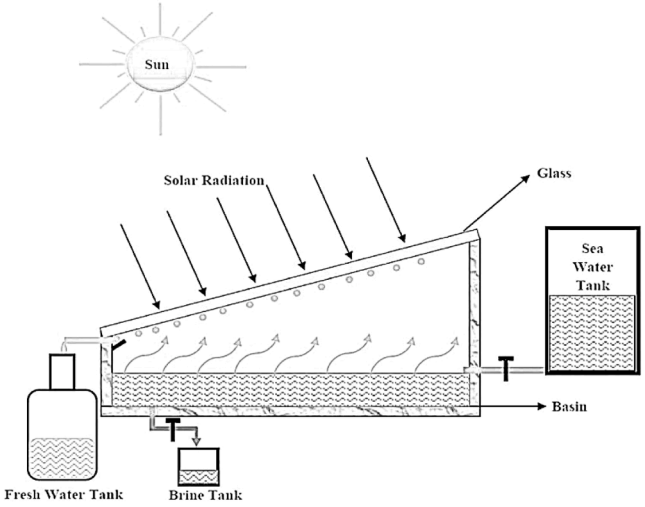


Fig. 1. Illustration of a basic solar-still system (adapted from Kapoor et al., 2023). The solar-still system uses solar radiation to heat the saltwater in the basin. The water evaporates and condenses on the glass cover, collecting as freshwater and flowing into the freshwater tank. The concentrated brine collects in the brine tank. This illustration showcases the process of solar-powered desalination.

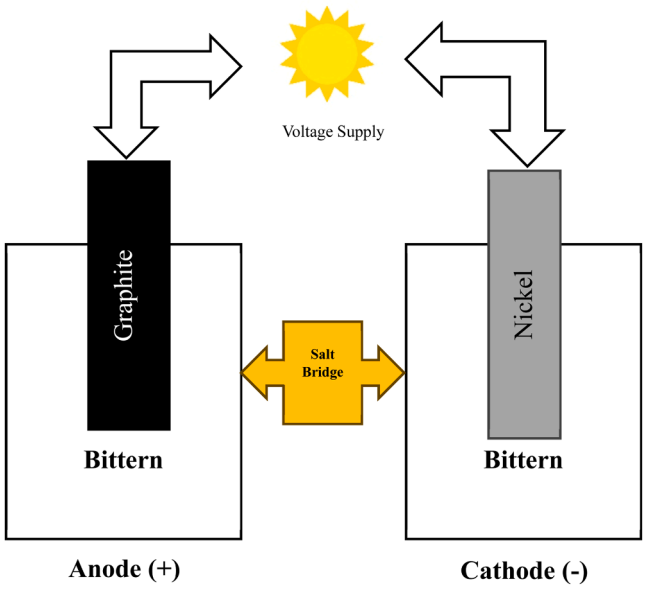


Fig. 2. Illustration of the solar electrochemical distillation process and its mechanism of action.

to drive the distillation process. The primary solar input during this process can be either through solar panels or photovoltaic cells, which collect solar energy and convert it into electricity. This process occurs actively in the second component, which is the electrochemical cell [21,22]. This cell features two electrodes – an anode and a cathode. The electrodes are separated by a membrane that allows ions to pass through while preventing the mixing of the anode and cathode compartments. When saline water, either seawater or brackish water, is introduced into the anode compartment, solar energy drives the electrochemical reaction. Simultaneously, positively charged ions, referred to as cations, move through the membrane towards the cathode, leaving behind the salts. The desalinated or fresh water is collected in the cathode compartment [19,21]. This process continues until sufficient water accumulates for use. Electrochemical cells use electricity to induce ionic movement, thereby improving the separation of salts and impurities from water.

2.1.2. Perceived advantages of SED

One of the most frequently cited advantages of SED is its high energy efficiency. Traditional desalination methods, such as reverse osmosis, require significant electrical energy to overcome osmotic pressure, which is the force applied to a solution to prevent solvent movement across a semipermeable membrane [12]. SED achieves high energy efficiency by harnessing solar energy, significantly reducing dependence on non-renewable energy sources and lowering operational costs. Furthermore, SED is considered most sustainable owing to its use of abundant renewable energy sources, which minimises greenhouse gas emissions [19,22], thereby, reducing the environmental impact associated with fossil fuel consumption.

Another advantage of SED is its economic viability. The incorporation of solar energy into the desalination process reduces energy expenditure, making SED a cost-effective approach [23]. While the initial cost of investing in solar technology is high, it is offset by the long-term savings in energy and maintenance costs. These enhance the economic viability of SED in large-scale desalination projects.

The SED approach is scalable and flexible. The system can be designed to scale to deal with projects of different capacities, ranging from small and decentralised units to large and centralised plants, specifically located in urban areas [19,22]. Hence, SED can be used to meet specific water desalination requirements of different regions and populations.

2.1.3. Potential applications

The use of SED in water desalination could be advantageous in various applications, including the supply of drinking water, agricultural irrigation and industrial applications. Huang et al. [24] describes a case study of a three-dimensional evaporator that applies the principles of solar electrochemical distillation to extract clean water from sea and wastewater. The fabricated Cu_xS-based solar water evaporator achieved a very high water evaporation rate of 1.96 kg m^{-2} h^{-1} and had an energy efficiency of 94.5 %. The evaporator also showed superior capability for multiphase purification of clean water. Table 1 presents a summary of SED's potential applications.

2.1.4. Challenges and prospects

Although SED is a promising approach that can facilitate the water desalination process, it faces several technical challenges that must be addressed before it can be fully effective. These include optimising the efficiency of solar absorbers, boosting the performance of electrochemical cells, and developing materials that can withstand difficult operating conditions [19].

Current estimates suggest that the initial capital costs for SED systems can be close to £0.019 per litre for passive solar stills. Depending on the selling price of distilled water, the payback period for these systems can range between 1.1–7.6 years. SED systems with active solar stills have much higher capital costs since they require extra equipment, such as pumps and controls. Consequently, the specific costs of SED systems with solar stills can vary based on the scale and the technology used [26]. Furthermore, the energy requirements for these systems vary based on the design and technology used. When used with passive solar stills, energy efficiency can range between 30 and 45 %. These systems are cost-effective and have low-carbon emissions since they use freely available solar energy to distil water. When used on multi-stage solar stills, they can achieve energy efficiencies of up to 53 % in scenarios where materials, such as paraffin infused with graphene nanocomposites, are used for thermal storage [27].

Typical permeate TDS levels achieved through SED depend on the design and quality of feed water. SED systems with passive solar stills can achieve permeate TDS levels ranging between 500 and 1000 ppm (parts per million). The value can be much lower and ranges between 300 and 500 ppm with active solar stills and below 300 ppm with multi-stage solar stills [28,29]. The recovery rates can vary depending on the system's design and the technology. SEDs with passive solar stills have recovery rates between 30 and 50 %. The values range between 50 and 70 % with active solar stills and can exceed 70 % with multi-stage solar stills [28,30]. Apart from the design, the recovery rate depends on the quality of the feed water and the environmental conditions.

If production of 1000 litres of distilled water is needed through passive solar still using a SED approach and the cost per litre is £0.019 with a total volume of 1000 litres:

Table 1
Potential applications of SED.

Application	Explanation
Supply of drinking water	SED can be used for the supply of safe and clean drinking water to communities living in water-scarce regions. By utilising local solar resources, SED systems produce fresh water sustainably, guaranteeing a reliable supply for daily consumption [19].
Agricultural irrigation	The availability of fresh water is important for crop production. This approach can provide an efficient solution for desalinating brackish water and making it suitable for irrigation. This method can improve agricultural productivity in arid regions, thereby, contributing to food security [25].
Industrial applications	Industries, especially those in coastal areas, usually need large quantities of water to facilitate different processes. SED can provide a reliable alternative for industrial water desalination, reducing the strain on fresh water resources and encouraging sustainable industrial practices [19].

Total cost = Cost per litre * Total volume

Total cost = 0.019 L * 1000 = £19

Using the payback period range of 1.1–7.6 years, the annual savings or revenue needed to achieve payback within this timeframe is calculated.

If the selling price of distilled water is £0.39 per litre:

Annual revenue from selling water = Selling price * Volume produced annually

If the production is 1000 litres per month:

Annual Volume = 1000 l/month * 12 = 12,000 l/year

= £0.39 * 12,000 = £4680 year

We also calculated the expected volume of fresh water generated based on recovery rates and the initial saline water input.

When starting with 1 m^3 (1000 litres) of saline water:

Recovery rate for passive solar stills (assuming an average of 40 %):
Fresh water volume Produced = (Initial volume) * (Recovery rate) = 1000 * 0.40 = 400 l

If the TDS levels are expected to be close to 500 ppm for passive solar stills, then for every litre of fresh water produced, it contains close to 500 mg of dissolved solids.

Continued research and development are important to overcome the technical challenges associated with the SED approach. Advancements in nanomaterials, technologies related to solar absorbers and electrochemical processes are essential for enhancing the efficiency and scalability of SED systems. This approach is largely governed by policies and regulations. SED adoption requires additional policy frameworks and regulations that support investments in renewable energy and sustainable desalination technologies [19]. Governments and international organisations can deploy SEDs by providing incentives, subsidies and regulatory support.

Overall, SED represents a primary development in the field of water desalination. By collecting solar energy and incorporating electrochemical processes, SED provides a sustainable, energy-efficient and cost-effective method for addressing global water scarcity. Although challenges exist, additional research, innovation and supportive policies can encourage the widespread adoption of SED. If well utilised, this approach can ensure a reliable supply of clean water for the imminent generation.

2.2. Integrated solar capacitive deionisation and hybrid systems

A promising emerging technology for water desalination is integrated solar capacitive deionisation (SCDI; Fig. 3). This method integrates solar energy with capacitive deionisation (CDI) to enable efficient and sustainable water desalination [31,32]. The system comprises two key components: a solar-energy collector and a CDI cell. The solar-energy collector captures and converts solar radiation into electrical energy, powering the CDI process.

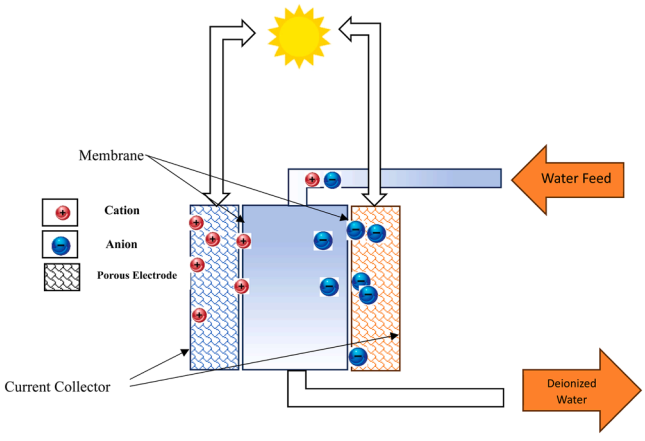


Fig. 3. Illustration of the capacitive deionization process drawing energy from the sun.

2.2.1. Mechanism

Integrated SCDI systems harness solar energy to power desalination through capacitive deionisation. The system comprises two main components: a solar-energy collector and a CDI cell. The solar-energy collector captures and converts solar radiation into electrical energy, driving the CDI process [31,32]. In the CDI cell, an electric field applied across two electrodes generates an electrostatic force, attracting and removing charged ions, including salts and impurities. As saline water flows between the electrodes, the ions are absorbed onto the electrode surfaces [31,32]. This process produces deionised water, commonly referred to as fresh water. Reversing the polarity regenerates the electrodes, releasing the absorbed ions into a separate water stream.

2.2.2. Advantages of SCDI

One of SCDI's key advantages is its high energy efficiency. Unlike conventional desalination methods that consume significant amounts of energy, SCDI uses renewable solar energy to power the desalination process. This minimises reliance on non-renewable energy sources, thereby lowering operational costs. SCDI's use of solar energy makes it an environmentally sustainable alternative to traditional desalination technologies [33,34]. By reducing fossil fuel consumption and greenhouse gas emissions, it supports environmental preservation and combats climate change.

As a cost-effective solution, the current estimates suggest that the SCDI systems can operate effectively with low salinity feeds (up to 35,000–45,000 ppm for seawater, 1000–10,000 ppm for brackish water and 5000–20,000 ppm for industrial wastewater) [35]. It can achieve this performance while maintaining recovery rates between 50 and 70 %, depending on the design. However, CDI as a singular technology is more economical for low-salinity feeds compared to seawater or high-salinity feeds [35].

Similar to solar electrochemical desalination, SCDI can be designed to meet the requirements of different capacities, ranging from small and decentralised units suitable for rural communities to large centralised plants utilised in urban areas [33,34]. The scalability and flexibility of this system guarantee that it can be adapted to meet the specific needs of different regions and populations.

2.2.3. Hybrid systems: CDI-HS

Recent advancements have enabled the development of CDI hybrid systems (CDI-HS), which integrate CDI with other desalination technologies such as reverse osmosis, membrane distillation, and electrodialysis [12] (Fig. 4). By leveraging the strengths of each technology, these hybrid systems improve performance and overcome the limitations of standalone systems. For instance, CDI-RO systems pre-treat feedwater, reducing osmotic pressure, improving RO efficiency, and extending membrane lifespan by minimising fouling [36]. In CDI-MD systems, CDI pre-treats feedwater to reduce fouling potential and enhance overall MD efficiency [37]. CDI-ED systems enhance ion selectivity and operational efficiency by removing specific ions either before or after the electrodialysis process [38]. Such hybrid systems reduce energy consumption while enhancing overall desalination performance.

2.2.4. Potential applications

SCDI and CDI-HS have proven effective in delivering safe and clean drinking water to communities in water-scarce regions. By harnessing local solar resources, these systems sustainably produce fresh water for daily consumption. They improve performance while reducing costs, enabling cities and communities to secure sustainable freshwater

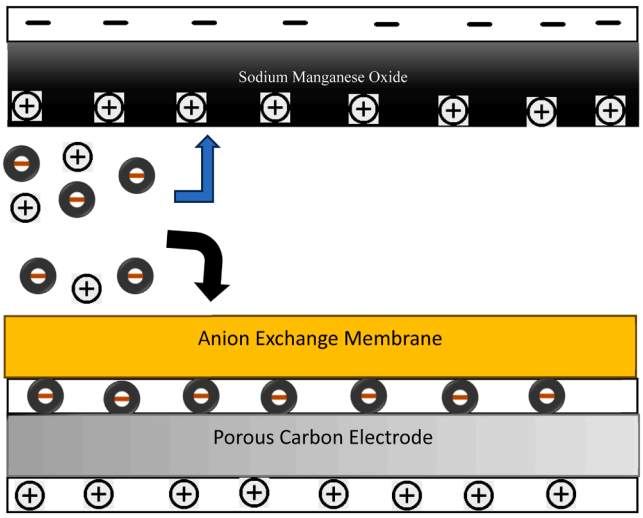


Fig. 4. Illustration of the mechanisms of action of capacitive deionization hybrid systems.

sources. Tan et al. [16] integrated a solar-energy supply with membrane CDI to facilitate salt removal from brackish water. Their results demonstrated that solar-powered membrane CDI (MCDI) water treatment, particularly when combined with energy recovery, was competitive with more established water treatment technologies. Fig. 5 shows a graphical representation of the process used to achieve 24-hour treatment of brackish water using the SCDI approach. The fresh water produced can be used for crop production and supply irrigation water to high-value crops [16]. Bales et al. [20] highlighted the use of solar energy in CDI as a low-cost approach for generating water for irrigation. Moreover, Bales et al. [21] found that a photovoltaic-powered operational-scale MCDI desalination system achieved an average power saving of 27 % when treating domestic wastewater for reuse. This can enhance agricultural productivity in arid regions and boost water and food security worldwide.

In industrial settings, these technologies provide a reliable supply of high-quality desalinated water while minimising operational costs and environmental impacts. Additionally, agricultural activities in arid regions rely on desalinated water for irrigation; therefore, SCDI and CDI-HS can improve efficiency and cost-effectiveness in agriculture, supporting food security. Tan et al. describe a case study integrating photovoltaic energy supply into a membrane capacitive deionization system. The resulting system can be used to treat brackish water in remote communities and generate drinking water [39]. It could also be used to provide irrigation water for high value crops.

2.2.5. Challenges and prospects

Although SCDI and CDI-HS represent significant advancements in desalination technology, they face challenges such as fouling on electrode surfaces caused by ion adsorption and accumulation of organic matter. Electrode fouling occurs when an inactive layer forms on the electrode surface, preventing effective ion adsorption and reducing system performance [35,40]. Organic matter accumulation on the electrode surface can further worsen fouling issues, affecting the overall efficiency and longevity of the system. These challenges require ongoing research into materials that reduce fouling and enhance system performance.

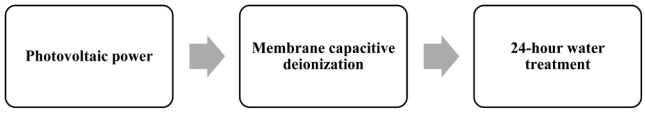


Fig. 5. Graphical representation of the process used to achieve 24-hour treatment of brackish water using the SCDI approach.

Additional research is needed to optimise recovery rates for different feed TDS levels, especially for higher salinity feeds where traditional CDI may struggle economically. Traditional CDI systems struggle with higher salinity feeds because of increased energy consumption and higher maintenance costs. Electrode fouling further reduces performance and increases operational costs in systems processing high-salinity feeds [35,41]. Supportive policies are needed to facilitate widespread adoption by incentivising research into innovative solutions that deal with these challenges.

These technologies are the primary development in the field of water desalination. They combine solar energy with CDI to offer a sustainable, energy-efficient and cost-effective solution to address global water scarcity. However, these methods have not been widely utilised across different industrial applications and several challenges remain. Additional research, innovation and supportive policies are needed to facilitate the widespread adoption of SCDI and CDI-HS. Proper utilisation of these approaches would ensure a reliable supply of clean water for future generations.

2.3. Artificial intelligence

The first traces of artificial intelligence (AI) emerged between 1950 and 1956 [42]. However, in recent years, the meaning of AI has revolutionised, transforming operations across different industries. Accurate forecasting and prediction of the distilled productivity has become a decisive issue to evaluate and compare the ability of different solar-still designs to supply distilled water without being costly and consuming too much time [43,44]. Researchers believe that AI can be integrated into water desalination processes as a revolutionary method to enhance efficiency, reduce costs and optimise operations.

2.3.1. Mechanism of action

AI involves various technologies, including machine learning, deep learning, neural networks and predictive analytics. Particularly, machine learning methods have become strong alternatives to traditional modelling techniques in different engineering disciplines [43]. These technologies can be incorporated into desalination processes to monitor, control and optimise different operational parameters. The main AI applications in desalination include predictive maintenance [45,46], process optimisation [46], energy management [45,47] and quality control [45,48]. Recent research has proposed two improved machine learning models – relevance vector machine (RVM) [49] and least squares support vector machine (LSSVM) [50] to predict the productivity of solar distillers [43]. Specifically, these improved machine learning models have been found to predict the distilled production of a modified stepped corrugated solar still with determination coefficient values of 0.999 and 0.982, and root mean square error values of 6.646 e-06 and 0.002, respectively [43]. Artificial neural networks (ANN) based models have also become popular in the recent times, having been used to model different engineering systems. ANNs use a valid dataset to model complex engineering problems. The models achieve this by simulating the relationship between the outputs and the inputs in the training process [43].

AI operation in water desalination systems functions in the following manner. Sensors and Internet of Things (IoT) devices collect real-time data on different operational parameters, such as temperature, pressure, water quality metrics and flow rates. In the data-processing phase, AI algorithms analyse the collected data to identify patterns, predict potential issues and optimise process parameters. During predictive maintenance, AI models predict potential equipment failures and maintenance needs, reducing downtime and increasing the lifespan of the desalination system. The optimisation process involves machine learning algorithms, optimising the operation of desalination plants by controlling variables to achieve maximum efficiency and reduce energy consumption [45–48]. The final step is quality control, in which AI

systems continuously monitor the water quality and ensure that the desalinated water meets the set health and safety standards. Fig. 6 illustrates the framework for integrating AI into the water desalination process.

2.3.2. Advantages of AI in water desalination

An advantage of using AI for water desalination is enhanced efficiency. This technology can significantly improve the efficiency of desalination processes by optimising the operational parameters in real-time. Machine learning algorithms can analyse large quantities of data to identify the most efficient settings, minimising energy consumption and operation costs [46]. Furthermore, AI techniques play an important role in modelling, optimising and controlling a wide range of applications, such as wastewater treatment [27–29]. This study followed the methodology of Aghdam et al. [27], which involved data collection, data normalisation, development of different AI models with pre-processed data to predict different parameters, model validation and assessment of the sensitivity of the target parameters towards the predictors. Nourani et al. [29] applied three different AI-based non-linear models, including a feed-forward neural network, an adaptive neuro-fuzzy inference system (ANFIS) and a support vector machine. Based on their results, the ANFIS model provided more effective outcomes than the single models. Sayed et al. [26] analysed the application of AI techniques for modelling, optimising and controlling desalination systems powered by renewable energy sources and noted potential efficiency benefits if the system was designed appropriately. However, they observed that the integration of these technologies into water desalination systems was challenging owing to unpredictable load demands and the intermittent nature of renewable energy sources. Nasr et al. [28] further noted that AI modelling and simulation suffered from a high level of nonlinearity, especially when multiple complex physical, chemical and biological processes are involved. Additionally, they noted the need to implement intelligent techniques that could integrate all working and design parameters involved in various possible scenarios.

Even though AI is not a desalination technology, it plays a central role in optimising existing technologies and improving the overall system performance. Thus, it has transformative potential in water desalination. As demonstrated in previous studies, AI has predominantly been applied in wastewater treatment plants. It can enhance efficiency, optimise energy use, predict maintenance needs and guarantee water quality. Furthermore, AI can address the main challenges of desalination and contribute to sustainable solutions for global water scarcity. Although challenges remain, continued research, innovation and supportive policies are needed to pave the way for the widespread adoption of AI in desalination. This will ensure a reliable and sustainable supply of clean water for the future.

Krishnan et al. [51] provides a case study of how artificial intelligence could be used to improve the water desalination process. For example, they describe the use of AI Sensing systems that are integrated with intelligent control methods in a nano-membrane plant. These sensors are designed to monitor key process variables such as membrane performance, quality of feedwater, and energy consumption. Monitoring these variables enhances the performance of the system.

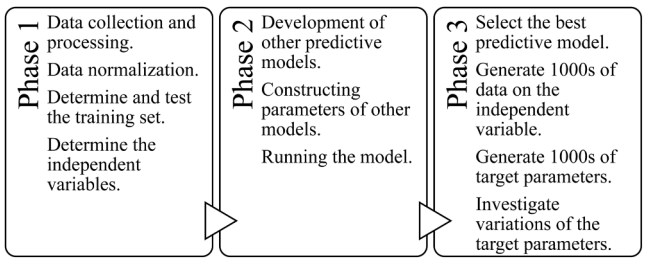


Fig. 6. Illustration of the general framework for AI integration in the water desalination process.

2.4. Uniform density desalination membranes

The development of Uniform Density Desalination Membranes (UDDMs) marks a significant advancement in desalination technology, offering solutions to long-standing efficiency challenges (Fig. 7). Desalination membranes are responsible for removing salt and other chemicals from water, a process that is important for purifying billions of gallons of water for agriculture, energy production and the provision of safe drinking water [52]. Desalination membranes operate by separating salts and impurities from water, producing clean water. While appearing straightforward, the desalination process involves complex mechanisms that continue to be the focus of scientific investigation. However, an important aspect of this mystery could be solved using UDDMs, which could reduce the costs involved in the production of clean water. Traditional desalination membranes often exhibit inconsistent densities and mass distributions, which adversely impact their performance [52]. Hence, membranes with uniform density at a nanoscale are key to increasing the water quantity that these membranes can produce.

2.4.1. Mechanism of action

UDDMs are engineered to maintain uniform density across the membrane surface, improving both performance and durability. Traditional membranes, such as those used in RO systems, frequently exhibit density variations. This variation leads to uneven flow distribution, increased fouling and reduced efficiency. UDDMs address these challenges by providing a uniform structure that ensures consistent water flow and salt rejection.

This mechanism often involves the following three steps – membrane fabrication, water flow and salt rejection and maintenance and durability. During membrane fabrication, UDDMs are manufactured using advanced techniques that ensure uniform density across the membrane [53,54]. This process is usually precise and controls the pore size and distribution, resulting in a highly consistent membrane structure. Saline water passes through the UDDM, enabling uniform water flow and efficient salt rejection. With uniform density, water flows evenly across the membrane surface, reducing the high flux areas that can lead to fouling. Salts and impurities are rejected more efficiently because of the consistent pore structure. The uniform membrane density reduces the risk of fouling and scaling, extends the membrane's lifespan and reduces maintenance requirements [53,54]. These improve the overall

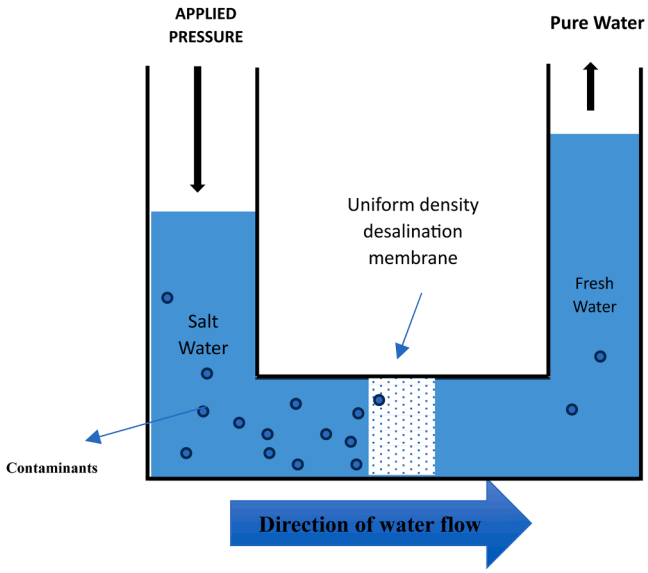


Fig. 7. Illustration of uniform density desalination membranes in action.

operational efficiency and cost-effectiveness of desalination.

2.4.2. Advantages of UDDMs

UDDMs offer significantly greater efficiency compared to traditional desalination membranes. A uniform density guarantees consistent water flow and salt rejection, reduces energy consumption and enhances the desalination process [52,53], thus, decreasing the operational costs and increasing the water production rate.

Fouling and scaling, common in traditional desalination processes, lead to frequent maintenance and shortened membrane lifespans. With the uniform density of UDDMs, the areas of high flux are minimised and the irregular flow distribution is managed [53]. This significantly reduces fouling and scaling, resulting in a more stable and reliable operation.

The enhanced efficiency of UDDMs results in substantial energy savings. By ensuring that the water flow is consistent and reducing the need for high pressures, these membranes reduce the energy requirements for desalination [52]. This innovation makes desalination systems more sustainable and economical, particularly in regions with high energy costs.

Furthermore, UDDMs provide superior salt rejection and contaminant removal capabilities, resulting in desalinated water of a higher quality [55]. The uniform pore structure guarantees that the smallest impurities are effectively removed, producing water that meets stringent health and safety standards.

2.4.3. Potential applications

UDDMs play a key role in providing reliable and efficient desalination of municipal water supplies. By enhancing the process and reducing desalination costs, UDDMs can help water-scarce regions secure a sustainable source of clean water. This technology can be applied to industries that require large quantities of high-quality water [52]. The membranes provide cost-effective and reliable solutions for industrial desalination. This ensures a consistent supply of desalinated water while reducing operation costs and environmental impact. Moreover, these membranes may be useful in agricultural irrigation. Agricultural activities in arid and semiarid regions often depend on desalinated water for irrigation and UDDMs can enhance the efficiency and cost-effectiveness of desalination for agricultural use, sustainable farming practices and food security [52].

Overall, UDDMs are an important technology that can prove useful in water desalination. This technology can provide better efficiency, reduce fouling and scaling, save energy and improve water quality, offering a promising solution for global water scarcity.

2.5. Solar-powered passive desalination

Recently, solar-powered passive desalination (SPPD) emerged as an innovative, energy-efficient and sustainable alternative to traditional methods (Fig. 8). According to a recent article by the Massachusetts Institute of Technology (MIT), such a system was developed by researchers at MIT and China. This system can provide >1.5 gallons of fresh drinking water per hour for every square metre of the solar collecting area [56]. The SPPD systems can potentially serve off-grid arid coastal areas as an efficient and low-cost water resource.

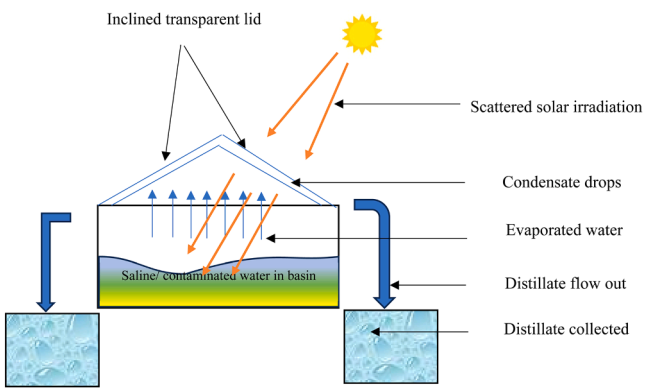


Fig. 8. Illustration of solar-powered passive desalination system.

2.5.1. Mechanism of action

SPDD uses solar energy to drive the desalination process without external power sources or complex mechanical systems. This technology involves solar stills, which use solar radiation to evaporate seawater and condense the vapour to produce fresh water. The system utilises numerous layers of flat solar evaporators and condensers, vertically lined up and topped with transparent aerogel insulation. A detailed description of the system is provided by Xu et al. [35].

Desalination involves three key processes – solar collection, evaporation and condensation. During solar collection, a solar still or another collection device is used to capture solar radiation and convert it into thermal energy. During evaporation, thermal energy heats the saline water in the still, resulting in evaporation [57,58], whereby, impurities and salts are left behind. During condensation, water vapour condenses on a cool surface and is collected as fresh water. This process is similar to the natural water cycle and is fully controlled by solar energy.

2.5.2. Advantages of SPPD

As SPPD relies entirely on solar energy, it is highly energy efficient and reduces the need for external power sources, thus, decreasing the operational costs and carbon footprint of the desalination process. The key to the system's efficiency lies in the fact that at each stage, the previously released heat is collected instead of being wasted to manner desalinate the water. Thus, the technology can achieve an overall efficiency of 385 % in converting solar energy into water evaporation energy [56]. The structure is essentially a multilayer solar still featuring a set of evaporating and condensing components, such as those used to distil liquids. It features flat panels that absorb heat and transfer it to a layer of water, which starts to evaporate. The vapour condenses on the next panel, allowing water to collect as the heat from the vapour condensation is passed to the next layer. After the vapour condenses on the surface, heat is released. In a typical condenser system, heat is lost to the environment [59]. However, in this multilayer evaporator, the released heat flows to the next evaporating layer, recycling the solar heat and enhancing the overall efficiency.

The simplicity of the SPPD system results in low operational and maintenance costs. Without moving parts or complex machinery, this system is easy to maintain and exhibits a long operational lifespan. Unlike traditional desalination systems, this technology does not result in the accumulation of salt or concentrated brine, which must be disposed of. This is because its configuration is free-floating, allowing any accumulated salt to be carried out at night through the wicking material and back into the seawater [56]. Furthermore, SPPD systems can be scaled to meet various requirements, ranging from small, decentralised units for individual households to larger systems for communities. This flexibility makes it accessible to remote and underserved regions where conventional desalination infrastructures may be impractical. Moreover, the system can be configured with floating panels on a saltwater body, such as an impoundment pond. Such a configuration can constantly and passively deliver fresh water through pipes to the shore if solar energy is available. It is possible to design a system to serve a single household, using a flat panel on a large shallow tank of seawater that is pumped or carried in.

Since SPPD systems are meant for small-scale applications in remote areas, their potential for commercialisation in large urban centres is limited by various factors. First, scaling up SPPD systems to meet the high water demands of large cities can be challenging. For solar to be collected extensively, adequate land area is needed, and this can be prohibitive in densely populated urban environments. These systems

may be further limited by production capacity. Current estimates indicate that every square meter can produce close to 1.5 gallons per hour. For large cities that need a lot of water, this production rate may not be sufficient unless bigger infrastructure and various units are installed. Furthermore, it is important to consider the economic viability of these systems before investing in them. Even though the initial costs for small family-sized systems can range close to £78, scaling these systems for larger applications can lead to higher capital costs because of additional materials and infrastructure requirements [56]. Moreover, its integration with existing infrastructure can often be challenging. Large cities depend on established water supply systems that may not easily integrate with new passive desalination technologies without requiring major changes in the design structure.

Even with these challenges, SPPD systems can be effectively used in specific urban scenarios, such as community gardens, emergency relief efforts during disasters or as additional sources of fresh water where there is no conventional infrastructure.

2.5.3. Potential applications

Notably, SPPD provides a reliable source of clean drinking water for water-scarce regions. Using local resources, these systems can ensure a sustainable supply of fresh water for daily consumption. Agriculture in arid regions often relies on desalinated water [60,61]. SPPD can produce high-quality water for irrigation, support sustainable farming practices and enhance food security. Access to clean water is critical in disaster-stricken areas or during emergencies. Portable SPPD units can be rapidly deployed to provide immediate fresh water supply, aid relief efforts and improve resilience [60]. Moreover, remote and off-grid locations often face significant challenges when accessing fresh water and SPPD systems offer a viable solution by providing a dependable water source without the need for complex infrastructure or external power.

2.6. Membrane distillation

Membrane distillation (MD) has emerged as a promising technology offering a novel desalination approach by combining the benefits of membrane technology and thermal processes (Fig. 9). This technique uses a separation process where a microporous hydrophobic membrane distinguishes between two aqueous solutions at different temperatures [62]. Due to its hydrophobic, the membrane prevents liquid mass transfer and creates a gas-liquid interface. This approach differs from RO, which is controlled by the osmotic pressure of the treated water. In contrast, MD is not limited by osmotic pressure, allowing for higher water recovery, even with bilge water containing elevated salinity [63].

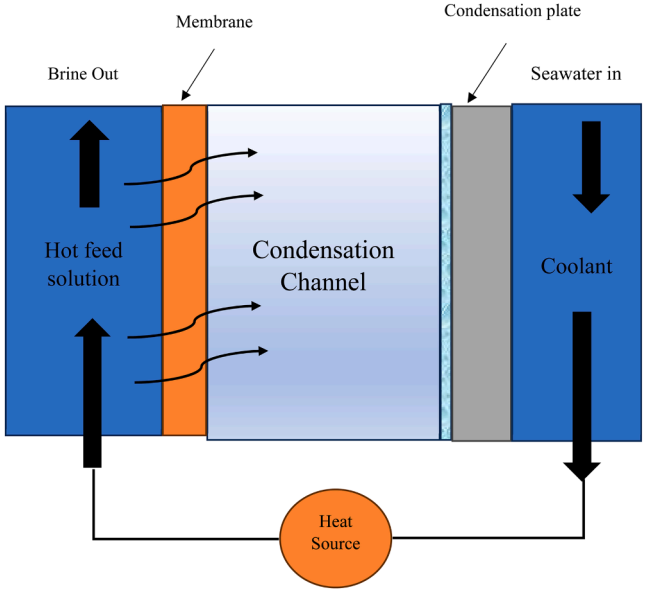


Fig. 9. Illustration of membrane distillation.

2.6.1. Mechanism

MD is a thermally driven separation process where only vapour molecules pass through a hydrophobic membrane, separating the feed solution from the permeate. The driving force behind MD is the vapour pressure difference created by the temperature gradient across the membrane [63]. MD consists of four processes: feed solution heating, vapour formation, membrane passage and condensation. In the feed solution heating stage, the saline feed solution is heated to create a temperature gradient across the hydrophobic membrane. In the vapour formation stage, the heated feed solution comes into contact with the membrane, allowing water vapour to form because of the temperature difference [64,65]. Membrane passage occurs when the hydrophobic membrane permits only water vapour to pass, blocking the liquid phase and dissolved salts. In the condensation stage, water vapour passes through the membrane and condenses on the cooler permeate side, producing desalinated water.

Different configurations of MD exist, including direct contact MD (DCMD) [66], air-gap MD (AGMD) [67], vacuum MD (VMD) [68] and sweeping gas MD (SGMD) [69]. In DCMD, the feed solution and permeate are in direct contact with the membrane. The temperature gradient across the membrane drives vapour transport. In AGMD, an air gap is introduced between the membrane and the condensation surface, reducing heat loss and improving thermal efficiency [66]. VMD requires the application of a vacuum on the permeate side, which increases the vapour pressure difference and enhances the mass-transfer rate [68]. In SGMD, a gas stream sweeps the permeate side and carries the vapour to an external condenser [69].

2.6.2. Advantages of MD

MD is a low-energy desalination process, especially when compared to traditional thermal methods. It operates at lower temperatures and can utilise low-grade or waste heat sources, thereby, significantly reducing energy consumption. MD achieves high salt rejection rates due to the hydrophobic nature of the membrane, which blocks dissolved salts and other non-volatile contaminants [64,65]. As a result, MD produces high-quality desalinated water.

Furthermore, MD systems are modular and can be scaled to meet various requirements, ranging from small, decentralised units for rural communities to large-scale installations in urban areas. This flexibility makes it adaptable to different water sources and demands. It can be effectively integrated with renewable energy sources, such as solar and geothermal energy, to enhance sustainability and reduce fossil fuel reliance. Particularly, solar-driven MD systems offer promising solutions for off-grid and remote locations [64]. The lower operating temperatures of MD reduce the risk of membrane fouling and scaling compared with high-temperature thermal desalination processes. This results in longer membrane life and lower maintenance costs.

2.6.3. Disadvantages and challenges

Despite its advantages, the MD system faces several challenges that need addressing. MD often shows lower recovery rates compared to other desalination technologies due to limitations in vapour flux and thermal efficiency. MD typically shows recovery rates between 50 and 90 % [70], whereas other desalination methods, like reverse osmosis, can achieve recovery rates as high as 98 %, particularly when integrated with other processes [12,71].

One key issue affecting MD systems is membrane wetting where liquid water penetrates the hydrophobic membrane pores, leading to decreased performance and higher operational costs. Studies have shown that membrane wetting happens at different rates depending on the membrane material and the operational conditions [72]. For example, constant wetting mode, where the isolated pores remain

wetted throughout the operation, can increase distillate salinity. An increased rate of wetting mode, where many pores wet over time, can majorly impact the performance of the desalination system [72].

Fouling is a major challenge for MD systems. As organic matter accumulates on the membrane surface, it can reduce efficiency and lifespan by obstructing vapour transport. Studies justify that fouling can result in a decline in permeate flux by up to 50 % within a few hours of operation. Such a reduction directly impacts the efficiency and lifespan of the MD system [73].

Currently, there are few commercially viable MD modules on the market; most existing systems are expensive or still in experimental phases [74,75]. Additionally, the high energy consumption of MD systems, for instance, the energy costs associated with maintaining the necessary temperature gradient for vapour transport, is a key barrier to their commercialisation [74,75].

These challenges require ongoing research into materials that reduce fouling and enhance performance while focusing on developing cost-effective commercial modules.

2.6.4. Potential applications

MD is a reliable source of clean drinking water for communities in water-scarce regions. By utilising local heat and renewable energy sources, MD systems ensure a sustainable supply of fresh water for daily consumption. Large amounts of high-quality water are required for various industrial processes. It can provide a cost-effective and reliable solution for industrial desalination, ensuring a consistent supply of desalinated water, while reducing operational costs and environmental impacts [64].

Agricultural activities in arid and semiarid regions often rely on desalinated water for irrigation. It improves the efficiency and cost-effectiveness of desalination for agricultural purposes, supports sustainable farming practices and enhances food security. MD can be used for wastewater treatment and recycling, providing an efficient method for recovering clean water from industrial and municipal wastewater streams [76]. Hence, it supports water conservation and sustainable water management practices.

2.7. Forward osmosis

Forward osmosis (FO) has emerged as an alternative desalination technology due to its low-energy requirements and high efficiency (Fig. 10). This process is a water separation technique that uses a semipermeable membrane and the natural energy of osmotic pressure to separate water from dissolved solutes [77]. Osmotic pressure drives water through the membrane while retaining all dissolved solutes on the other side.

2.7.1. Mechanism of action

FO involves a membrane-based separation process that uses the natural osmotic pressure gradient between a concentrated draw solution and a feed solution (e.g. seawater) to drive water through a semi-permeable membrane. Unlike RO, FO relies on natural osmotic pressure rather than hydraulic pressure, making it more energy efficient [77].

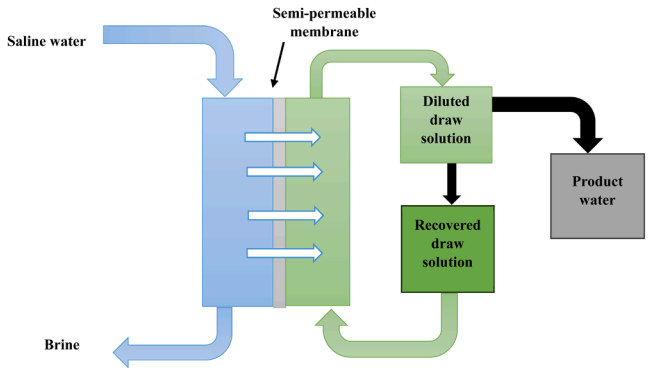


Fig. 10. Illustration of the mechanism of action of forward osmosis.

It works through four key processes – osmotic pressure gradient, water flux, dilution of the draw solution and separation of fresh water. During the osmotic pressure gradient, the FO process begins with two solutions separated by a semipermeable membrane – a draw solution with high osmotic pressure and a feed solution with low osmotic pressure [78]. During water flux, water molecules from the feed solution naturally diffuse through the membrane towards the draw solution because of the osmotic pressure gradient, leaving salts and impurities behind. The draw solution is diluted as it absorbs water from the feed solution. Finally, the diluted draw solution is processed to separate fresh water from the draw solutes, which can be reused in a closed-loop system [78].

2.7.2. Advantages of FO

FO offers several advantages over traditional desalination methods. Since FO relies on natural osmotic pressure rather than mechanical pressure, it is inherently more energy efficient than traditional desalination methods. This significantly reduces energy consumption and operational costs, especially for high-salinity feedwater. Studies have shown that FO can save up to 20 % of energy at 70 % recovery and 40 % energy at 90 % recovery compared to competing processes [79].

Furthermore, FO membranes experience less fouling and scaling than RO membranes. Lower operational pressure reduces the physical stress on the membrane, leading to a longer membrane lifespan and reduced maintenance costs [80]. Studies have shown that FO membranes have a lower fouling propensity, which reduces the maintenance and operation costs [81]. FO can handle a wide range of feed water sources, including seawater, brackish water and wastewater. This versatility makes it suitable for various applications, ranging from municipal water supply to industrial wastewater treatment. FO has been effectively implemented in municipal water supply systems to generate potable water from different sources, including impaired water streams [82]. This makes it a suitable option for addressing water scarcity in urban areas.

FO systems can achieve high water recovery rates, which makes them efficient for desalination in water-scarce regions [83]. The process can be optimised to maximise water extraction from feed solutions and reduce the water volume. FO systems can achieve water recovery rates of close to 95 %; a value higher than traditional RO systems that achieve rates close to 75 % [84]. Moreover, FO systems can be easily integrated with renewable energy sources, such as solar or geothermal energy, further enhancing their sustainability and reducing the carbon footprint of desalination. For instance, the Lawrence Berkeley National Laboratory (LBNL) and the California Resources Corporation (CRC) are developing an integrated water treatment system using FO powered by solar energy. The system is designed to treat high-salinity wastewater from oil and gas fields [85].

2.7.3. Challenges and prospects

Despite its advantages, FO faces several technical challenges, including the development of effective draw solutes, membrane fouling and the need for efficient methods to separate fresh water from draw solutions. Research and innovation are essential for addressing these issues and improving FO performance.

The choice of the draw solution is critical for FO efficiency. Ideal draw solutes should have high osmotic pressure and low reverse solute flux and should be easily separable from the product water. Draw solutions that are commonly used include salts and sugars. However, the recovery and regeneration of these solutions can be complex and costly. Besides, these solutions face challenges in achieving osmotic pressure and reverse solute flux [86]. Continued research on novel draw solutes is vital for the advancement of FO technology [87]. Investigating alternative draw solutes, such as organic compounds or ionic liquids, could

provide higher osmotic pressures with lower reverse flux characteristics. Designing hybrid draw solutions that bring together various solutes could optimise performance.

It is challenging to regenerate the draw solution after it has been diluted. The key challenges relate to energy consumption and operational complexity. Advanced methods are needed to separate fresh water from diluted draw solutions, as this would improve the overall efficiency of the system. For example, thermal methods used to regenerate draw solutes need a lot of energy input, which increases the operation costs of FO systems [87]. Future research could investigate energy-efficient regeneration techniques. Researchers could explore innovative separation technologies, such as membrane distillation or advanced filtration methods, which could reduce energy requirements during regeneration.

Advancements in membrane materials are key to enhancing FO efficiency and durability. The cost of membranes can be significant, even though they are generally less expensive than RO membranes because of lower operational pressures. FO membranes can cost approximately £787 per square meter or more [88]. Research on new membrane materials that present higher permeability, selectivity and resistance to fouling could facilitate future development of FO systems. Furthermore, research could focus on developing scalable manufacturing processes for advanced membranes, which could reduce costs.

Even though FO systems have high recovery rates, they are likely to be lower compared to other technologies, such as RO, under certain conditions. Optimising the system's design and operation parameters is important for maximising the recovery rates. Future research should focus on optimising system configurations and operational strategies to enhance water recovery rates in FO systems. Besides, research into advanced control algorithms or machine learning techniques could enhance real-time monitoring and adjustment of the operation parameters of these systems.

3. Discussion on unresolved issues from current review articles

Despite significant advancements in desalination technologies, numerous unresolved issues hinder their widespread adoption and efficiency. By analysing current review articles and research data, this study offers a comprehensive overview of the challenges and potential solutions.

3.1. Efficiency, scalability and material durability

Key unresolved issues with SED include efficiency, scalability, material durability, data and research gaps. SED systems harness solar energy to drive the electrochemical processes and facilitate desalination. However, the efficiency of solar energy capture and conversion remains a major challenge. Existing solar absorbers and photovoltaic cells have limited efficiencies, which affect the overall performance of SED systems. Additionally, scaling-up SED systems for large-scale applications poses technical and economic challenges. Moreover, the materials used in these systems, especially for electrodes and membranes, must be able to withstand tough operational conditions, including high salinity and temperature fluctuations. The durability and longevity of these materials are important for ensuring the system's reliability and cost-effectiveness. Current research has examined advanced materials with better durability and performance that can be used in this system; nevertheless, no definitive solution exists. Studies have indicated that improving the efficiency of solar absorbers can improve the general performance of SED systems [89]. Increasing the efficiency of solar cells from 20 % to 25 % could potentially result in a similar increase in desalination efficiency. Furthermore, research has been conducted using new materials, such as graphene-based electrodes [90], which have shown great promise, yet require additional development to address durability-related issues.

3.2. Energy recovery and electrode fouling

Several issues related to energy recovery and electrode fouling in the SCDI system remain unresolved. The SCDI system integrates photovoltaic cells with CDI, thus providing a sustainable solution for desalination. However, energy recovery during desalination continues to be a concern. Capturing and reusing energy efficiently, particularly during regeneration, is important for improving the overall energy efficiency of the system. Furthermore, electrode fouling results from the accumulation of organic and inorganic contaminants. This persistent issue reduces the efficiency of ion removal and intensifies the need for system maintenance. Therefore, developing anti-fouling materials and self-cleaning mechanisms is important [91]. However, data and research on SCDI systems are scarce. Recent research has focused on improving the energy-recovery mechanisms of these systems. For example, utilising an energy storage device, such as a super capacitor, can enhance energy-recovery efficiency by approximately 30 % [92]. Moreover, the development of anti-fouling coatings for electrodes, such as those containing titanium dioxide nanoparticles, could reduce fouling by 40 % in laboratory tests.

3.3. Limited water output and weather dependency

Limited water output and weather dependency are the main unresolved issues in SPPD systems. SPPD technologies use solar energy to evaporate and condense water, mimicking the natural water cycle. These systems generate relatively low quantities of water compared with other desalination technologies. Therefore, the key challenge is increasing the rate of water production while maintaining low-energy consumption. Moreover, these systems depend on the weather conditions, particularly the availability of sunlight, implying that cloudy days and night-time can be challenging and reduce the efficiency of the system. These limitations can affect the reliability and continuous operation of the systems in areas with unpredictable weather patterns. Similar to other technologies, researchers have proposed innovations to enhance the operation of SPPD systems, some of which involve the use of advanced solar collectors and materials that store heat. Incorporating phase change materials (PCMs) is estimated to increase the water production rate by approximately 50 %. This is possible through the storage of solar energy during the day and its subsequent release at night [93]. Alternatively, hybrid systems that use solar energy and other renewable energy sources, such as wind or geothermal energy, can address weather dependency issues affecting this technology.

3.4. System integration, cost and complexity

The primary unresolved issues associated with CDI-HS are system integration, cost and complexity. These hybrid systems combine CDI with other desalination technologies, such as electro dialysis or RO, to improve efficiency. This integration introduces new challenges related to compatibility, control and optimisation. Hence, it is important to develop methods to ensure the seamless operation of the hybrid system and maximise the benefits of every component of the technology. The complex nature of these systems raises cost concerns [94]. As multiple components are required, the system's general cost increases, rendering it less accessible across several regions. Cost issues further render this technology less applicable in low-income regions. These issues require a large amount of data and research to optimise the control algorithms for CDI-HS. Additionally, integrated modules that combine CDI and RO into one unit must be created, as they have demonstrated the ability to reduce the complexity and costs of the hybrid system. Pilot projects have been implemented in arid regions [94] and have proven promising; however, deployment in large-scale setups remains a concern.

The primary unresolved issues with MD are membrane wetting, scaling, heat loss, and energy efficiency. MD uses hydrophobic membranes to separate water vapours from salts and impurities. This process

is significantly affected by membrane wetting and scaling, which reduce the system's efficiency and lifespan [95]. Therefore, creating membranes that can resist wetting and scaling while maintaining high permeability is important. Recent advancements in membrane materials, such as super hydrophobic coatings and nanocomposite membranes, have shown the potential to minimise wetting and scaling by approximately 60 %. Concerning heat loss and energy efficiency, MD processes suffer from significant heat loss, which decreases their overall efficiency. To make MD a suitable desalination technology for large-scale use, finding ways to reduce heat loss and enhance thermal efficiency is important. The incorporation of heat recovery systems, such as MED, can enhance the energy efficiency of MD by approximately 40 % [96].

3.5. Draw solution development and membrane fouling

FO faces challenges related to draw solution development and membrane fouling. Its efficiency is heavily dependent on the properties of the draw solutions. Identifying draw solutes that can provide high osmotic pressure, low reverse solute flux and easy separation from the product water remains a challenge. Recent studies have focused on creating new draw solutes, such as thermolytic and magnetic nanoparticles, that can be easily separated from the product water. These draw solutes can increase the osmotic pressure by approximately 40 % and decrease the reverse solute flux by 30 % [97]. Concerning membrane fouling, FO membranes are highly susceptible to fouling, which can reduce water flux and system efficiency. Combining anti-fouling membranes with better cleaning protocols is important to maintain the FO systems' performance. Progress in membrane surface modifications, such as the inclusion of anti-fouling coatings, has shown the potential to reduce fouling by 50 % [98]. Table 2 summarises the unresolved issues and justifies the benefits of addressing them.

Table 2
Unresolved issues according to current review articles.

Technology	Unresolved Issues	Details	Justification
SED	Efficiency and scalability	Limitations in solar-energy capture and conversion; challenges in scaling up for large-scale applications.	Improving solar absorber efficiency from 20 % to 25 % can increase desalination efficiency proportionally.
	Material durability	Need for advanced materials that withstand high salinity and temperature fluctuations.	Graphene-based electrodes show promise, however, require further development.
SCDI and CDI-HS	Energy recovery system integration	Efficient capture and reuse of energy during the regeneration phase. Ensuring compatibility, control and optimisation when integrating multiple systems.	Integrating super capacitors can improve energy-recovery efficiency by up to 30 %. Optimising control algorithms can enhance system efficiency by up to 20 %.
	Electrode fouling	Accumulation of organic and inorganic contaminants on the electrodes reduces efficiency.	Anti-fouling coatings with titanium dioxide nanoparticles reduce fouling by 40 % in laboratory tests.
SPPD	Limited water output	Relatively low water production rate compared to other technologies.	Integrating phase change materials can increase water production by up to 50 %.
	Weather dependency	Efficiency is significantly reduced on cloudy days and at night.	Hybrid systems combining solar with wind or geothermal energy can mitigate weather dependency.
	Cost and complexity	High initial costs and maintenance owing to multiple components and advanced control systems.	Integrated CDI-RO modules show promise of reducing complexity and costs.
MD	Membrane wetting and scaling	Developing membranes that resist wetting and scaling while maintaining high permeability.	Super hydrophobic coatings and nanocomposite membranes reduce wetting and scaling by up to 60 %.
	Heat loss and energy efficiency	Minimising heat loss and enhancing thermal efficiency.	Integrating heat recovery systems can improve energy efficiency by up to 40 %.
	Selectivity and efficiency	Achieving high selectivity and efficiency in ion removal while maintaining low-energy consumption.	Optimising electrode configuration and operating conditions can enhance selectivity and efficiency.
FO	Draw solution development	Finding draw solutes with high osmotic pressure, low reverse solute flux and easy separation.	Novel draw solutes, such as thermolytic and magnetic nanoparticles, increase osmotic pressure by up to 40 %.
	Membrane fouling	Developing anti-fouling membranes and effective cleaning protocols.	Anti-fouling coatings reduce fouling by 50 %.

4. Recommendations regarding the unresolved issues

Despite significant advancements in desalination technologies, several unresolved issues continue to hinder their widespread adoption and efficiency. Ongoing research and innovation are essential for addressing these challenges and enhancing their performance, scalability and cost-effectiveness. Emerging desalination technologies, such as SED, SCDI, SPPD, CDI-HS, MD, electrochemical processes and FO, could play a vital role in addressing global water scarcity and ensuring a sustainable supply of clean water for future generations.

The current landscape of the new desalination technologies presents both opportunities and challenges. This paper provides comprehensive recommendations for the unresolved issues identified in these technologies. These recommendations are supported by recent research data and aim to enhance the performance, scalability and sustainability of desalination processes.

4.1. Recommendations

4.1.1. Improve the capturing of solar energy and the conversion process efficiency

The findings highlighted above underscore the importance of improving the efficiency of solar absorbers and photovoltaic cells for the new desalination technologies to be used widely [62]. Current research efforts should concentrate on developing advanced materials and technologies that can enhance solar-energy conversion rates. A 5 % increase in the efficiency of photovoltaic cells could result in an equivalent increase in the efficiency of the desalination process [93].

4.1.2. Develop durable and high-performance materials

Extensive research is needed to develop advanced materials, such as graphene-based electrodes, capable of addressing durability and performance issues in SED systems. Their design must ensure that they can withstand high salinities and temperature fluctuations. Graphene-based electrodes show great promise in enhancing the durability and performance of desalination systems. They could increase the lifespan of electrodes by up to 50 % and improve the efficiency of the desalination

process by >20 % [90].

4.1.3. Enhance the energy recovery mechanisms

Developing efficient energy-recovery systems, such as those incorporating super capacitors, can significantly enhance the energy efficiency of the SCDI and other desalination systems. This would reduce energy consumption during the regeneration phase. Studies cited in this review suggest that integrating super capacitors would enhance energy recovery by approximately 30 % [91,92]. Super capacitors can store and reuse energy, leading to significant energy savings in water desalination systems. Hence, enhancing energy-recovery mechanisms is imperative.

4.1.4. Implement anti-fouling mechanisms

A primary method for addressing electrode fouling in SCDI systems is developing anti-fouling materials and self-cleaning mechanisms. This can increase the efficiency of ion removal from brackish water and reduce maintenance costs. The use of titanium dioxide nanoparticles in anti-fouling coatings has been shown to minimise fouling by 40 % during laboratory tests. Developing such coatings can extend the lifespan of the electrodes and enhance system efficiency [92].

4.1.5. Increase the water production rate

Integrating advanced solar collectors and heat storage materials, such as PCMs, can increase the amount of water produced by the SPPD systems. This ensures continuous operation even during low sunlight. Recent studies have shown that PCMs can increase water production by approximately 50 % [89]. They can store solar energy during the day and release it at night, ensuring a steady water output.

4.1.6. Create hybrid energy systems

The reliability of SPPD systems in areas with variable weather patterns can be improved by combining solar energy with other renewable energy sources, such as wind and geothermal energy. Hybrid systems using both solar and wind energy can reduce downtime and enhance overall efficiency by approximately 40 % [93]. They provide an incessant supply of energy and boost the performance of the SPPD systems.

4.1.7. Optimise system integration and control

Creating better control algorithms and integrated modules that use CDI along with other desalination technologies, such as RO, can enhance the system's performance and compatibility. Better control algorithms enhance system efficiency by approximately 20 % [94]. Integrated CDI-RO modules have been researched and show promise of reducing the complexity and cost of water desalination systems. Consequently, CDI-HS systems are more suitable for large-scale applications.

4.1.8. Reduce initial costs and complexity associated with desalination systems

This review demonstrates a need to invest in the development of integrated modules and streamlined designs. This can reduce the initial costs and complexity of CDI-HS systems. This method can make water desalination systems more accessible and easier to maintain. Modules that use both CDI and RO in a single unit have lowered costs by approximately 30 % for pilot projects [94]. Moreover, adopting simple designs guarantees system efficiency and reduces maintenance requirements.

4.1.9. Develop advanced membranes

Future research should focus on creating super hydrophobic and nanocomposite membranes that can deal with membrane wetting and scaling. These membranes provide high permeability and resistance to fouling and reduce wetting and scaling by approximately 60 % [96]. These advancements have significantly enhanced the efficiency and lifespan of MD systems.

4.1.10. Enhance thermal efficiency

Integrating heat recovery systems, such as MED, can reduce heat loss and enhance the thermal efficiency of MD processes. It renders MD more suitable for large-scale desalination. The research cited in this review shows that heat recovery systems can enhance the energy efficiency of MD processes by approximately 40 % [96]. These systems are designed to capture and reuse heat, thereby, reducing the overall energy consumption of MD processes.

4.1.11. Enhance the materials used in electrodes

The research and design of better electrode materials, such as graphene and carbon nanotubes, can boost the durability and performance of electrochemical desalination systems. They exhibit superior performances owing to high conductivity and resistance to potential corrosion. According to previous research, graphene and carbon nanotubes can improve the electrode lifespan by over 50 % and increase ion removal with an estimated efficiency of over 30 % [97].

4.1.12. Optimise selectivity and efficiency

Adjusting the configuration and operating conditions of the electrodes can boost the selectivity and efficiency of ion removal during electrochemical desalination. This process optimisation can balance energy consumption and optimise performance. Optimising the electrode design can boost efficiency by up to 25 % [97]. Furthermore, optimising the operating conditions for certain feed water compositions could further enhance performance.

4.1.13. Create effective draw solutions

Research is needed for new types of draw solutes, such as thermolytic and magnetic nanoparticles. An in-depth examination of these solutions could enhance the efficiency of FO processes. These new solutes can provide high osmotic pressure, low reverse solute flux and easy separation from the product water. Specifically, they can increase the osmotic pressure by over 40 % and reduce the reverse solute flux by over 30 % [97].

4.1.14. Create anti-fouling membranes

Investing in the development of anti-fouling membranes and improving cleaning protocols can alleviate membrane fouling issues in FO systems. These membranes could provide a high-water flux and resistance to contaminants. The anti-fouling coatings showed a 50 % decrease in the fouling rates. Using membrane materials of better quality and modifying the surface of the materials can boost the performance and lifespan of FO membranes.

4.2. Economic and policy implications

Effective adoption and implementation of the newly developed technologies needs a detailed understanding of economic and policy frameworks. This subsection presents the cost-benefit analysis, funding mechanisms, and economic incentives that are key to encouraging wider adoption.

4.2.1. Cost-Benefit analysis

Newly discovered technologies such as SED and FO promise major long-term savings by reducing the consumption of energy and lowering the costs of operation. However, the cost of acquiring these technologies remains high and surpasses that of traditional methods like RO. Even though the initial cost of acquiring the materials and designs is high, the expense is offset by the prolonged lifespans and fewer maintenance needs. Secondly, these technologies reduce energy consumption by 20–40 %, yielding major savings over time. This advantage is most evident in areas with high costs of energy. Finally, decreased reliance on fossil fuels and lower greenhouse gas emissions increases the overall value proposition of these systems by aligning with global sustainability goals. Governments and industries should review these benefits against

the initial high cost of acquisition to assess long-term viability, particularly in water-scarce areas.

4.2.2. Funding mechanisms

Large-scale deployment of desalination technologies would require funding. Potential funding mechanisms may include government grants, public-private partnerships (PPPs), international aid and development funds, and green bonds. National and local governments could provide grants to cater for the initial cost of acquisition. Government entities could collaborate with private industries to pool resources together for infrastructure development. Successful PPPs could see the operation of newly developed desalination systems scaled. International aid organisations like the World Bank could also offer loans or grants to support projects in developing nations. Lastly, green bonds could attract investors interested in sustainability initiatives.

4.2.3. Economic incentives

Policy makers could encourage adoption of emerging desalination systems by implementing economic incentives. One incentive would be tax credits or breaks for companies that adopt energy-efficient desalination technologies. Besides, subsidies could reduce the financial burden placed on early adopters. Lastly, policy makers could design preferential tariffs for industries or municipalities by integrating advanced desalination methods.

5. Conclusion and future research

Worldwide water scarcity requires desalination technologies to evolve and improve continuously. This review discussed new desalination technologies, such as SED, SCDI, SPPD, CDI-HS, MD and FO. Each technology has unique advantages. However, it faces several challenges that could hinder its widespread adoption and efficiency.

Based on these unresolved issues, we recommend several improvement strategies. Improving the efficiency of solar absorbers and photovoltaic cells is necessary for maximising energy conversion in SED systems. A focused increase in efficiency could lead to major enhancements in the general performance of the desalination process. A comprehensive analysis of solar-energy efficiency should be performed to identify the most efficient and cost-effective solutions for SED systems. Such studies should compare different photovoltaic materials, solar absorbers and concentrator designs under different environmental conditions. Furthermore, they should investigate the long-term performances and degradation rates of these components. Therefore, future research should focus on evaluating the long-term durability and performance of advanced materials, such as graphene-based electrodes, under different desalination conditions. Extensive field tests should be performed to examine the degradation rates, facilitate better mechanical stability and improve the efficiency of ion removal from new materials over extended periods. Researchers can investigate different energy-recovery methods and their integration into SCDI systems to improve energy efficiency. Moreover, they can explore the use of super capacitors, energy-recovery turbines and other storage devices to model energy flow and identify the most effective recovery techniques.

It is important to address the identified issues through focused research and implementation of recommended strategies to boost the performance, scalability and sustainability of these technologies. A strong possibility exists that this can enhance the efficiency of the desalination processes and reduce the operation costs and environmental impact. These strategies will make clean water more accessible worldwide. Hence, researchers must develop and test new anti-fouling coatings and self-cleaning mechanisms for the SCDI electrodes. They can perform experiments using different types of nanomaterials and surface treatments to reduce fouling. Additionally, they should assess the effectiveness of these coatings in laboratory and field settings. Furthermore, future studies should examine the use of novel materials and designs to increase the water output from the SPPD systems. Researchers should study the integration of advanced solar collectors, PCMs and other heat storage solutions to boost efficiency and water production. It is worth exploring the potential of hybrid systems using solar and other renewable energy sources to enhance the reliability of the SPPD systems. Researchers can analyse different hybrid systems under different weather conditions and model energy integration and storage capabilities for continuous operation.

The gradual use of advanced desalination technologies, along with innovative research and policy frameworks, could prove useful in addressing the global water crisis. This is because the responsibility of guaranteeing reliable and sustainable clean water for future generations requires collaboration across scientific, industrial and government sectors. Leveraging the advantages of each desalination technology and the challenges likely to arise can pave the way for a more water-secure world.

For FO technology, it is necessary to explore new draw solutes that provide high osmotic pressure with low reverse solute flux characteristics. This could enhance the overall efficiency of FO desalination systems. Researchers can develop and test novel draw solutes and can focus on thermolytic and magnetic nanoparticles as possible draw solutes. Laboratory and field tests can be performed to evaluate their performance and feasibility. Studies on anti-fouling membrane development can be used to research effective cleaning protocols capable of reducing fouling in FO systems. Researchers can experiment with different membrane materials, surface modifications and cleaning techniques and evaluate their effectiveness in maintaining high water flux and fouling resistance.

Future research could focus on creating membranes that deal with wetting and scaling issues in MD processes, especially on super hydrophobic and nanocomposite membranes with high permeability and resistance to fouling. Researchers should conduct laboratory and field tests to evaluate their performance. These membranes are important for improving the lifespan and efficiency of desalination systems. Studies investigating methods to reduce heat loss and enhance the thermal efficiency of MD processes are needed. These studies can examine the integration of heat recovery systems into different water desalination systems. They can model the heat flow and identify areas that require improvement.

As noted in the CDI-HS systems, optimising control algorithms and integrating them with other desalination technologies could enhance performance and compatibility. Optimisation could make these systems more suitable for application in large-scale settings. Further studies should focus on developing and validating better control algorithms and integrated modules for CDI-HS systems. Researchers can test different control strategies and integration designs to enhance the performance and compatibility of different water desalination systems. Additionally, they can conduct real-time monitoring and adaptive control experiments. Studies can focus on cost-reduction strategies, specifically identifying and implementing cost-reduction strategies for CDI-HS systems to make them more accessible. Furthermore, they can investigate cost-effective materials, manufacturing processes and streamlined systems. The economic feasibility of these projects should be tested through pilot projects.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data statement

This is a review article, and no original data were generated or analysed during this research. All data referred to in this review are available from the respective articles and publications that were reviewed. Readers can access the data by referring to the sources cited within this article.

CRediT authorship contribution statement

Anwur Alenezi: Writing – review & editing, Writing – original draft, Validation, Supervision, Conceptualization. **Yousef Alabaiadly:** Writing – review & editing, Resources, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

[1] D. Zhang, M.S. Sial, N. Ahmad, A.J. Filipe, P.A. Thu, M. Zia-Ud-Din, A.B. Caleiro, Water scarcity and sustainability in an emerging economy: a management perspective for future, Sustainability 13 (2021) 144, https://doi.org/10.3390/su13010144.

[2] World Wildlife Fund Water scarcity available online: https://www.worldwildlife.org/threats/water-scarcity/(accessed on 24 May 2024).

[3] N. Dhakal, S.G. Salinas-Rodriguez, J. Hamdani, A. Abushaban, H. Sawalha, J. C. Schippers, M.D. Kennedy, Is desalination a solution to freshwater scarcity in developing countries? Membranes 12 (2022) 381, https://doi.org/10.3390/membranes12040381.

[4] S.A. El-Agouz, A.R. Abd Elbar, A.M. Aboghazala, M. Shahin, M.Y. Zakaria, K. K. Esmaeil, M.E. Zayed, Comprehensive parametric analysis, sizing, and performance evaluation of a tubular direct contact membrane desalination system driven by heat pipe-based solar collectors, Energy Convers. Manag. 274 (2022) 116437, https://doi.org/10.1016/j.enconman.2022.116437.

[5] S.M. Shalaby, F.A. Hammad, M.E. Zayed, Current progress in integrated solar desalination systems: prospects from coupling configurations to energy conversion and desalination processes, Process Saf. Environ. Prot. 178 (2023) 494–510, https://doi.org/10.1016/j.psep.2023.08.058.

[6] J.J. Feria-Díaz, M.C. López-Méndez, J.P. Rodríguez-Miranda, L.C. Sandoval-Herazo, F. Correa-Mahecha, Commercial Thermal Technologies For Desalination of Water from Renewable Energies: A State of the Art Review, 9, Processes, 2021, p. 262, https://doi.org/10.3390/pr9020262.

[7] S.M. Shalaby, M.E. Zayed, F.A. Hammad, A.S. Menesy, A.R.A. Elbar, Recent advances in membrane distillation hybrids for energy-efficient process configurations: technology categorization, operational parameters identification, and energy recovery strategies, Process Saf. Environ. Prot. 190 (2024) 817–838, https://doi.org/10.1016/j.psep.2024.07.098.

[8] I.S Al-Mutaz, MSF challenges and survivals, Desalination Water Treat. 177 (2020) 14–22, https://doi.org/10.5004/dwt.2020.24908.

[9] S. Azimibavil, A. Jafarian Dehkordi, Dynamic simulation of a multi-effect distillation (MED) process, Desalination 392 (2016) 91–101, https://doi.org/10.1016/j.desal.2016.04.004.

[10] Y. Li, X. Chen, Y. Xu, Y. Zhuo, G. Lu, Sustainable thermal-based desalination with low-cost energy resources and low-carbon footprints, Desalination 520 (2021) 115371, https://doi.org/10.1016/j.desal.2021.115371.

[11] M. Doninelli, E. Morosini, G. Gentile, L. Putelli, G. Di Marcoberardino, M. Binotti, G. Manzolini, Thermal desalination from rejected heat of power cycles working with CO2-based working fluids in CSP application: a focus on the MED technology, Sustain. Energy Technol. Assess. 60 (2023) 103481, https://doi.org/10.1016/j.seta.2023.103481.

[12] M.M. Aboelmaaref, J. Zhao, M.E. Zayed, Y. Li, L. Gu, A.A. Askalany, M. Ghazy, A. S. Alsaman, E.S. Ali, Design and dynamic numerical modeling of a hybrid reverse osmosis/adsorption-based distillation system driven by solar dish stirling engine for enhanced performance and waste heat recovery, Process Saf. Environ. Prot. 180 (2023) 324–338, https://doi.org/10.1016/j.psep.2023.10.012.

[13] M. Kerper, Optimizing Reverse Osmosis Recovery Rates and Membrane Health, EAI Water, 2024.

[14] A. Alhawsawi, M.E. Zayed, E. Moustafa, E. Banoqitah, A.H. Elsheikh, Hybridizing solar dish stirling power system with single-effect desalination for sustainable electricity and freshwater Co-generation: mathematical modeling and performance evaluation, Case Stud. Therm. Eng. 45 (2023) 102997, https://doi.org/10.1016/j.csite.2023.102997.

[15] H. Yuan, J. Liu, X. Zhang, L. Chen, Q. Zhang, L. Ma, Recent advances in membrane-based materials for desalination and gas separation, J. Clean. Prod. 387 (2023) 135845, https://doi.org/10.1016/j.jclepro.2023.135845.

[16] M.A. Tony, Central composite design optimization of bismarck dye oxidation from textile effluent with Fenton's reagent, Appl. Water. Sci. 10 (2020) 108, https://doi.org/10.1007/s13201-020-01192-5.

[17] M.A. Tony, H.A. Nabwey, Recent advances in solar still technology for solar water desalination, Appl. Water. Sci. 14 (2024) 147, https://doi.org/10.1007/s13201-024-02188-1.

[18] S. Shoeibi, M. Saemian, M. Khiadani, H. Kargarsharifabad, S. Ali Agha Mirjalily, Influence of PV/T waste heat on water productivity and electricity generation of solar stills using heat pipes and thermoelectric generator: an experimental study and environmental analysis, Energy Convers. Manag. 276 (2023) 116504, https://doi.org/10.1016/j.enconman.2022.116504.

[19] S. Cheng, Y. Li, Z. Yu, Y. Sun, B. Jin, Z. Mao, In situ thermal and electricity utilization of photovoltaic devices by membrane distillation and electrochemical advanced oxidation for desalination and degradation of wastewater, Adv. Sustain. Syst. 5 (2021) 2000278, https://doi.org/10.1002/adsu.202000278.

[20] M. Tao, (Invited) Electrochemistry for sustainable Solar photovoltaics, Meet. Abstr (2019) 2063, https://doi.org/10.1149/MA2019-01/43/2063. *MA2019-01*.

[21] D. Amiruddin, D. Mahajan, D. Fang, W. Wang, P. Wang, B.S. Hsiao, A facile ultrapure water production method for electrolysis via multilayered photovoltaic/membrane distillation, Energies (Basel) 16 (2023) 5765, https://doi.org/10.3390/en16155765.

[22] M. Tao, Review—Electrochemistry for sustainable solar photovoltaics, ECS J. Solid. State Sci. Technol. 9 (2020) 125007, https://doi.org/10.1149/2162-8777/abd376.

[23] S. Licht, O. Chitayat, H. Bergmann, A. Dick, H. Ayub, S. Ghosh, Efficient STEP (Solar Thermal Electrochemical Photo) production of hydrogen – an economic assessment, Int. J. Hydrogen Energy 35 (2010) 10867–10882, https://doi.org/10.1016/j.ijhydene.2010.07.028.

[24] W. Huang, P. Su, Y. Cao, C. Li, D. Chen, X. Tian, Y. Su, B. Qiao, J. Tu, X. Wang, Three-dimensional hierarchical CuxS-based evaporator for high-efficiency multifunctional solar distillation, Nano Energy 69 (2020) 104465, https://doi.org/10.1016/j.nanoen.2020.104465.

[25] E. Espinoza Márquez, G.M. Soto Zarazúa, J.de J Pérez Bueno, Prospects for the use of electrooxidation and electrocoagulation techniques for membrane filtration of irrigation water, Environ. Process 7 (2020) 391–420, https://doi.org/10.1007/s40710-020-00439-2.

[26] K.R. Ranjan, S.C. Kaushik, Economic feasibility evaluation of solar distillation systems based on the equivalent cost of environmental degradation and high-grade energy savings, Int. J. Low-Carbon Technol. 11 (2016) 8–15, https://doi.org/10.1093/ijlct/ctt048.

[27] H.B. Bacha, A.S. Abdullah, F.A. Essa, Z.M. Omara, Energy, exergy, economic, and environmental prospects of solar distiller with three-vertical stages and thermo-storing material, Processes 11 (2023) 3337, https://doi.org/10.3390/pr11123337.

[28] H. Manchanda, M. Kumar, A comprehensive decade review and analysis on designs and performance parameters of passive solar still, Renew. Wind. Water. Sol. 2 (2015) 17, https://doi.org/10.1186/s40807-015-0019-8.

[29] A.H. Mohammed, A.N. Shmroukh, N.M. Ghazaly, A.E. Kabeel, Active solar still with solar concentrating systems, review, J. Therm. Anal. Calorim. 148 (2023) 8777–8792, https://doi.org/10.1007/s10973-023-12285-z.

[30] S.R. Akkala, A. Kumar Kaviti, Advanced design techniques in passive and active tubular solar stills: a review, Environ. Sci. Pollut. Res. 29 (2022) 48020–48056, https://doi.org/10.1007/s11356-022-20664-6.

[31] C. Tan, C. He, W. Tang, P. Kovalsky, J. Fletcher, T.D. Waite, Integration of photovoltaic energy supply with membrane capacitive deionization (MCDI) for salt removal from Brackish Waters, Water Res. 147 (2018) 276–286, https://doi.org/10.1016/j.watres.2018.09.056.

[32] Q. Wu, D. Liang, S. Lu, H. Wang, Y. Xiang, D. Aurbach, E. Avraham, I. Cohen, Advances and perspectives in integrated membrane capacitive deionization for water desalination, Desalination 542 (2022) 116043, https://doi.org/10.1016/j.desal.2022.116043.

[33] L. Wang, J.E. Dykstra, S. Lin, Energy Efficiency of capacitive deionization, Environ. Sci. Technol. 53 (2019) 3366–3378, https://doi.org/10.1021/acs.est.8b04858.

[34] C. Zhang, L. Wu, J. Ma, A.N. Pham, M. Wang, T.D. Waite, Integrated flow-electrode capacitive deionization and microfiltration system for continuous and energy-efficient brackish water desalination, Environ. Sci. Technol. 53 (2019) 13364–13373, https://doi.org/10.1021/acs.est.9b04436.

[35] M. Liu, M. He, J. Han, Y. Sun, H. Jiang, Z. Li, Y. Li, H. Zhang, Recent advances in capacitive deionization: research progress and application prospects, Sustainability 14 (2022) 14429, https://doi.org/10.3390/su142114429.

[36] W. Zhang, B. Ding, Y. Wu, X. Li, F. Cheng, X. Zhang, Performance optimization and comparison of two hybrid regeneration methods for absorption air-conditioning system, Solar Energy 249 (2023) 81–97, https://doi.org/10.1016/j.solener.2022.11.013.

[37] C. Skuse, A. Gallego-Schmid, A. Azapagic, P. Gorgojo, Can emerging membrane-based desalination technologies replace reverse osmosis? Desalination 500 (2021) 114844 https://doi.org/10.1016/j.desal.2020.114844.

[38] A. Narayanan, P. T, N.K. K, P.B. M, M. Padaki, Electrochemical elegance: a review on capacitive deionization for fluoride and arsenic mitigation in Brackish water, Electrochim. Acta 503 (2024) 144875, https://doi.org/10.1016/j.electacta.2024.144875.

[39] C. Tan, C. He, W. Tang, P. Kovalsky, J. Fletcher, T.D. Waite, Integration of photovoltaic energy supply with membrane capacitive deionization (MCDI) for salt removal from Brackish waters, Water Res. 147 (2018) 276–286, https://doi.org/10.1016/j.watres.2018.09.056.

[40] B.L. Hanssen, S. Siraj, D.K.Y. Wong, Recent strategies to minimise fouling in electrochemical detection systems, Rev Anal Chem 35 (2016) 1–28, https://doi.org/10.1515/revac-2015-0008.

[41] Y. Jiang, L. Jin, D. Wei, S.I. Alhassan, H. Wang, L. Chai, Energy consumption in capacitive deionization for desalination: a review, Int. J. Environ. Res. Public Health 19 (2022) 10599, https://doi.org/10.3390/ijerph191710599.

[42] J. Fleck, Development and establishment in artificial intelligence. The Question of Artificial Intelligence, Routledge, 1987. ISBN 978-0-429-50533-1.

[43] M.E. Zayed, V.P. Katekar, R.K. Tripathy, S.S. Deshmukh, A.H. Elsheikh, Predicting the yield of stepped corrugated solar distiller using kernel-based machine learning models, Appl. Therm. Eng. 213 (2022) 118759, https://doi.org/10.1016/j.applthermaleng.2022.118759.
[44] W.H. Alawee, L.A. Al-Haddad, H.A. Dhahad, S.A. Al-Haddad, Predicting the cumulative productivity of a solar distillation system augmented with a tilted absorber panel using machine learning models, J. Eng. Res. (Ponta Grossa) (2024), https://doi.org/10.1016/j.jer.2024.01.007.
[45] S.S. Ray, R.K. Verma, A. Singh, M. Ganesapillai, Y.-N. Kwon, A holistic review on how artificial intelligence has redefined water treatment and seawater desalination processes, Desalination 546 (2023) 116221, https://doi.org/10.1016/j.desal.2022.116221.
[46] M. Abd Elaziz, M.E. Zayed, H. Abdelfattah, A.O. Aseeri, E.M. Tag-eldin, M. Fujii, A. H. Elsheikh, Machine learning-aided modeling for predicting freshwater production of a membrane desalination system: a long-short-term memory coupled with election-based optimizer, Alex. Eng. J. 86 (2024) 690–703, https://doi.org/10.1016/j.aej.2023.12.012.
[47] M.A.E. Mohamed, S.M.R. Mohamed, E.M.M. Saied, M. Elsisi, C.-L. Su, H.A. Hadi, Optimal energy management solutions using artificial intelligence techniques for photovoltaic empowered water desalination plants under cost function uncertainties, IEEE Access 10 (2022) 93646–93658, https://doi.org/10.1109/ACCESS.2022.3203692.
[48] E.T. Sayed, A.G. Olabi, K. Elsaid, M. Al Radi, C. Semeraro, M.H. Doranehgard, M. E. Eltayeb, M.A. Abdelkareem, Application of artificial intelligence techniques for modeling, optimizing, and controlling desalination systems powered by renewable energy resources, J. Clean. Prod. 413 (2023) 137486, https://doi.org/10.1016/j.jclepro.2023.137486.
[49] M.R. Jabbari, M.M. Dorafshan, S. Eslamian, Chapter 22 - relevance vector Machine (RVM), in: S. Eslamian, F. Eslamian (Eds.), Handbook of Hydroinformatics, Elsevier, 2023, pp. 365–384. ISBN 978-0-12-821285-1.
[50] Y. Cai, S. Hao, X. Wen, H. Li, X. He, L. Chen, J. Ren, Short-term power prediction by using least square support vector machine with variational mode decomposition in a photovoltaic system, IEEE Access 11 (2023) 143486–143500, https://doi.org/10.1109/ACCESS.2023.3343103.
[51] A. Krishnan, T. Sundaram, B. Nagappan, Y. Devarajan, Bhumika integrating artificial intelligence in Nanomembrane systems for advanced water desalination, Results Eng. 24 (2024) 103321, https://doi.org/10.1016/j.rineng.2024.103321.
[52] Breed, E. Desalination breakthrough could lead to cheaper water filtration available online: https://news.utexas.edu/2020/12/31/desalination-breakthrough-could-lead-to-cheaper-water-filtration/ (accessed on 9 July 2024).
[53] V. Freger, G.Z. Ramon, Polyamide desalination membranes: formation, structure, and properties, Prog. Polym. Sci. 122 (2021) 101451, https://doi.org/10.1016/j.progpolymsci.2021.101451.
[54] Y.G. Yan, W.S. Wang, W. Li, K.P. Loh, J. Zhang, A graphene-like membrane with an ultrahigh water flux for desalination, Nanoscale 9 (2017) 18951–18958, https://doi.org/10.1039/C7NR07193K.
[55] H.Y. Yang, Z.J. Han, S.F. Yu, K.L. Pey, K. Ostrikov, R. Karnik, Carbon nanotube membranes with ultrahigh specific adsorption capacity for water desalination and purification, Nat. Commun. 4 (2013) 2220, https://doi.org/10.1038/ncomms3220.
[56] Chandler, D.L. Simple, solar-powered water desalination available online: https://news.mit.edu/2020/passive-solar-powered-water-desalination-0207 (accessed on 9 July 2024).
[57] O. Bait, Direct and indirect solar–powered desalination processes loaded with nanoparticles: a review, Sustain. Energy Technol. Assess. 37 (2020) 100597, https://doi.org/10.1016/j.seta.2019.100597.
[58] S. Kumar, M. Kumar, S. Chowdhury, B.S. Rajpurohit, J.K. Randhawa, Environmental concerns and long-term solutions for solar-powered water desalination, J. Clean. Prod. 345 (2022) 131180, https://doi.org/10.1016/j.jclepro.2022.131180.
[59] F.M. Abed, M.S. Kassim, M.R. Rahi, Performance improvement of a passive solar still in a water desalination, Int. J. Environ. Sci. Technol. 14 (2017) 1277–1284, https://doi.org/10.1007/s13762-016-1231-9.
[60] G. Baggio, J. Adamowski, V.J. Hyde, M. Qadir, Small-scale desalination and atmospheric water provisioning systems in water-scarce vulnerable communities: status and perspectives, Int. J. Water. Resour. Dev. 40 (2024) 686–717, https://doi.org/10.1080/07900627.2023.2273475.
[61] Z.Y. Ho, R. Bahar, C.H. Koo, Passive solar stills coupled with fresnel lens and phase change material for sustainable solar desalination in the tropics, J. Clean. Prod. 334 (2022) 130279, https://doi.org/10.1016/j.jclepro.2021.130279.
[62] S.A. El-Agouz, M.E. Zayed, A.M. Abo Ghazala, A.R.A. Elbar, M. Shahin, M. Y. Zakaria, K.K. Ismaeil, Solar thermal feed preheating techniques integrated with membrane distillation for seawater desalination applications: recent advances, retrofitting performance improvement strategies, and future perspectives, Process Saf. Environ. Prot. 164 (2022) 595–612, https://doi.org/10.1016/j.psep.2022.06.044.
[63] P. López-Porfiri, S. Ramos-Paredes, P. Núñez, P. Gorgojo, Towards the technological maturity of membrane distillation: the MD module performance curve, NPJ Clean. Water. 6 (2023) 1–9, https://doi.org/10.1038/s41545-023-00234-0.
[64] S. Jeon, H. Park, J. Yeo, S. Yang, C.H. Cho, M.H. Han, D.K. Kim, Desalination via a new membrane capacitive deionization process utilizing flow-electrodes, Energy Environ. Sci. 6 (2013) 1471–1475, https://doi.org/10.1039/C3EE24443A.
[65] Y. Wimalasiri, L. Zou, Carbon nanotube/graphene composite for enhanced capacitive deionization performance, Carbon N Y 59 (2013) 464–471, https://doi.org/10.1016/j.carbon.2013.03.040.
[66] R. Ullah, M. Khraisheh, R.J. Esteves, J.T. McLeskey, M. AlGhouti, M. Gad-el-Hak, H. Vahedi Tafreshi, Energy efficiency of direct contact membrane distillation, Desalination 433 (2018) 56–67, https://doi.org/10.1016/j.desal.2018.01.025.
[67] S. Kalla, S. Upadhyaya, K. Singh, Principles and advancements of Air gap membrane distillation, Rev. Chem. Eng. 35 (2019) 817–859, https://doi.org/10.1515/revce-2017-0112.
[68] M.A.E.-R. Abu-Zeid, Y. Zhang, H. Dong, L. Zhang, H.-L. Chen, L. Hou, A comprehensive review of vacuum membrane distillation technique, Desalination. 356 (2015) 1–14, https://doi.org/10.1016/j.desal.2014.10.033.
[69] I.A. Said, T. Chomiak, J. Floyd, Q. Li, Sweeping gas membrane distillation (SGMD) for wastewater treatment, concentration, and desalination: a comprehensive review, Chem. Eng. Process. Process Intensif. 153 (2020) 107960, https://doi.org/10.1016/j.cep.2020.107960.
[70] I. Chimanlal, L.N. Nthunya, C. Quist-Jensen, H. Richards, Membrane distillation crystallization for water and mineral recovery: the occurrence of fouling and its control during wastewater treatment, Front. Chem. Eng. (2022) 4, https://doi.org/10.3389/fceng.2022.1066027.
[71] W.N.A. Wan Osman, N.I. Mat Nawi, S. Samsuri, M.R. Bilad, Y. Wibisono, E. Hernández Yáñez, J. Md Saad, A review on recent progress in membrane distillation crystallization, ChemBioEng Rev. 9 (2022) 93–109, https://doi.org/10.1002/cben.202100034.
[72] A.L. McGaughey, A.E. Childress, Wetting indicators, modes, and trade-offs in membrane distillation, J. Memb. Sci. 642 (2022) 119947, https://doi.org/10.1016/j.memsci.2021.119947.
[73] N. AlSawaftah, W. Abuwatfa, N. Darwish, G. Husseini, A comprehensive review on membrane fouling: mathematical modelling, prediction, diagnosis, and mitigation, Water (Basel) 13 (2021) 1327, https://doi.org/10.3390/w13091327.
[74] A. Hussain, A. Janson, J.M. Matar, S. Adham, Membrane distillation: recent technological developments and advancements in membrane materials, Emergent. Mater. 5 (2022) 347–367, https://doi.org/10.1007/s42247-020-00152-8.
[75] A. Ali, M.M. Agha Shirazi, L. Nthunya, R. Castro-Muñoz, N. Ismail, N. Tavajohi, G. Zaragoza, C.A. Quist-Jensen, Progress in module design for membrane distillation, Desalination (2024) 581, https://doi.org/10.1016/j.desal.2024.117584.
[76] H. Julian, N. Nurgirisia, G. Qiu, Y.-P. Ting, I.G. Wenten, Membrane distillation for wastewater treatment: current trends, challenges and prospects of dense Membrane distillation, J. Water. Process. Eng. 46 (2022) 102615, https://doi.org/10.1016/j.jwpe.2022.102615.
[77] B. Peñate, L. García-Rodríguez, Current trends and future prospects in the design of seawater reverse osmosis desalination technology, Desalination 284 (2012) 1–8, https://doi.org/10.1016/j.desal.2011.09.010.
[78] S. Kook, C.D. Swetha, J. Lee, C. Lee, T. Fane, I.S. Kim, Forward osmosis membranes under null-pressure condition: do hydraulic and osmotic pressures have identical nature? Environ. Sci. Technol. 52 (2018) 3556–3566, https://doi.org/10.1021/acs.est.7b05265.
[79] O. Bakajin, J. Klare, E. Desormeaux, C. Benton, J. Lalikos, A. Rinella, A. May, Water and Energy Savings Using Innovative Forward Osmosis Systems for Irrigation and Indirect Potable Reuse, California Energy Commission, 2021.
[80] S.M. Ali, J.E. Kim, S. Phuntsho, A. Jang, J.Y. Choi, H.K. Shon, Forward osmosis system analysis for optimum design and operating conditions, Water Res. 145 (2018) 429–441, https://doi.org/10.1016/j.watres.2018.08.050.
[81] X. Hao, S. Gao, J. Tian, S. Wang, H. Zhang, Y. Sun, W. Shi, F. Cui, New insights into the organic fouling mechanism of an in situ Ca 2+ modified thin film composite forward osmosis membrane, RSC Adv. 9 (2019) 38227–38234, https://doi.org/10.1039/C9RA06272F.
[82] L. Zhu, C. Ding, T. Zhu, Y. Wang, A review on the forward osmosis applications and fouling control strategies for wastewater treatment, Front. Chem. Sci. Eng. 16 (2022) 661–680, https://doi.org/10.1007/s11705-021-2084-4.
[83] J. Wang, X. Liu, Forward osmosis technology for water treatment: recent advances and future perspectives, J. Clean. Prod. 280 (2021) 124354, https://doi.org/10.1016/j.jclepro.2020.124354.
[84] R. Wang, J. Li, C. Xu, X. Xu, F. Tang, M. Huang, Integrating reverse osmosis and forward osmosis (RO-FO) for printing and dyeing wastewater treatment: impact of FO on water recovery, Environ. Sci. Pollut. Res. 30 (2023) 92495–92506, https://doi.org/10.1007/s11356-023-28853-7.
[85] PROJECT PROFILE: lawrence Berkeley National Laboratory (Solar Desalination) available online: https://www.energy.gov/eere/solar/project-profile-lawrence-berkeley-national-laboratory-solar-desalination (accessed on 1 November 2024).
[86] Y. Xu, Y. Zhu, Z. Chen, J. Zhu, G. Chen, A comprehensive review on forward osmosis water treatment: recent advances and prospects of membranes and draw solutes, Int. J. Environ. Res. Public Health 19 (2022) 8215, https://doi.org/10.3390/ijerph19138215.
[87] Q. Long, Y. Jia, J. Li, J. Yang, F. Liu, J. Zheng, B. Yu, Recent advance on draw solutes development in forward osmosis, Processes 6 (2018) 165, https://doi.org/10.3390/pr6090165.
[88] A. El-Gendi, H. Abdallah, A. Amin, Economic study for blend membrane production, Bull. Natl. Res. Cent. 45 (2021) 126, https://doi.org/10.1186/s42269-021-00584-0.
[89] T. Arunkumar, R. Sathyamurthy, D. Denkenberger, S.J. Lee, Solar distillation meets the real world: a review of Solar stills purifying real wastewater and seawater, Environ. Sci. Pollut. Res. 29 (2022) 22860–22884, https://doi.org/10.1007/s11356-022-18720-2.
[90] B.H. Nguyen, V.H. Nguyen, Promising applications of graphene and graphene-based nanostructures, Adv. Nat. Sci: Nanosci. Nanotechnol. 7 (2016) 023002, https://doi.org/10.1088/2043-6262/7/2/023002.

[91] S. Liang, M. Li, J. Cao, K. Zuo, Y. Bian, K. Xiao, X. Huang, Integrated ultrafiltration–capacitive-deionization (UCDI) for enhanced antifouling performance and synchronous removal of organic matter and salts, Sep. Purif. Technol. 226 (2019) 146–153, https://doi.org/10.1016/j.seppur.2019.05.085.
[92] I.A. Said, N. Fuentes, Q. Li, Energy recovery in electrified capacitive deionization systems for wastewater treatment and desalination: a comprehensive review, Chem. Eng. Process. Process Intensif. 178 (2022) 109030, https://doi.org/10.1016/j.cep.2022.109030.
[93] T. Arunkumar, D. Parbat, S.J. Lee, Comprehensive review of advanced desalination technologies for solar-powered all-day, all-weather freshwater harvesting systems, Renew. Sustain. Energy Rev. 199 (2024) 114505, https://doi.org/10.1016/j.rser.2024.114505.
[94] Z.S. Alotaibi, K.N. Alharbi, Y. Alharbi, M.S. Almoiqli, Innovative pilot plant capacitive deionization for desalination brackish water, Appl. Water. Sci. 14 (2024) 26, https://doi.org/10.1007/s13201-023-02083-1.
[95] P. Wang, W. Cheng, X. Zhang, Q. Liu, J. Li, J. Ma, T. Zhang, Membrane scaling and wetting in membrane distillation: mitigation roles played by humic substances, Environ. Sci. Technol. 56 (2022) 3258–3266, https://doi.org/10.1021/acs.est.1c07294.
[96] E. Ali, J. Orfi, H. AlAnsary, J.-G. Lee, A. Alpatova, N. Ghaffour, Integration of multi effect evaporation and membrane distillation desalination processes for enhanced performance and recovery ratios, Desalination 493 (2020) 114619, https://doi.org/10.1016/j.desal.2020.114619.
[97] D. Zhao, S. Chen, C.X. Guo, Q. Zhao, X. Lu, Multi-functional forward osmosis draw solutes for seawater desalination, Chin. J. Chem. Eng. 24 (2016) 23–30, https://doi.org/10.1016/j.cjche.2015.06.018.
[98] N. Akther, S.M. Ali, S. Phuntsho, H. Shon, Surface modification of thin-film composite forward osmosis membranes with polyvinyl alcohol–Graphene oxide composite hydrogels for antifouling properties, Desalination 491 (2020) 114591, https://doi.org/10.1016/j.desal.2020.114591.